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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

A MACHINE LEARNING AND GIS-BASED APPROACH FOR SPATIAL PREDICTION OF SOIL FERTILITY IN SUNYANI, GHANA

BY

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DECLARATION

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ABSTRACT

Soil fertility is a critical determinant of agricultural productivity and food security, yet in many parts of Ghana, including the Sunyani Municipality, farmers lack timely, site-specific information to guide sustainable land management. Traditional soil assessment methods, while accurate, are often costly, time-consuming, and limited in scope, creating barriers to efficient decision-making. This study presents a Machine Learning and Geographic Information Systems (GIS)-based approach for developing a Soil Fertility Index (SFI) to spatially predict and classify soil fertility in Sunyani. Using soil nutrient data alongside environmental variables such as the Normalized Difference Vegetation Index (NDVI) and Digital Elevation Model (DEM), a Random Forest Regressor was trained and optimized to generate fertility scores on a continuous scale. The model achieved high predictive accuracy, with strong validation results confirmed through cross-validation and low error margins. A web-based GIS application, FertiTrack, was developed to provide interactive soil fertility maps, real-time predictions, and actionable recommendations for farmers and stakeholders. The system categorizes soils into fertility classes (Excellent, Good, Average, Poor), offering both precision and interpretability. Findings demonstrate that integrating machine learning with geospatial technologies significantly enhances soil fertility monitoring, enabling cost-effective, scalable, and user-friendly solutions for precision agriculture. Beyond its technical achievements, the project contributes to sustainable land management, food security, and climate-smart agriculture, aligning with global goals such as Zero Hunger (SDG 2), Climate Action (SDG 13), and Life on Land (SDG 15).

DEDICATION

We dedicate this work first to the Almighty God, whose guidance, wisdom, and strength enabled us to complete this project successfully.

To our families, for their unwavering love, encouragement, and sacrifices throughout our academic journey.

To our lecturers and mentors, whose knowledge, patience, and guidance inspired us to pursue excellence.

And finally, to the hardworking farmers of Ghana and beyond, whose resilience and dedication to feeding nations motivated us to develop this model in the hope of contributing to sustainable agriculture and improved livelihoods.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Soil fertility is a critical factor in sustaining crop productivity and meeting the food demands of the world's growing population. Assessing the fertility status of soil allows for the adoption of targeted agricultural practices to enhance crop yields. However, national-scale soil property maps in Ghana remain scarce.

In Ghana, agriculture holds paramount importance for both national development and the sustenance of its population. However, the country struggles to achieve the potential yields for staple crops. Notably, the reported yields of maize, soybean, and rice fall considerably below the potential yields (Ulzen et al., 2023).

In the Ghanaian context, agriculture remains a vital pillar of the economy. The sector contributes nearly 20% to the national GDP and employs about 45% of the total labor force, particularly in rural areas (GSS, 2020). Despite its significance, Ghana continues to face major challenges in agricultural productivity. The yields of major staple crops such as maize, rice, cassava, and soybean are often well below their potential. For instance, while the attainable yield for maize under optimal conditions can exceed 6 tons per hectare, the national average is about 1.8 tons per hectare (Ragasa et al., 2014). This persistent yield gap is largely attributable to poor soil fertility among other agronomic constraints.

Soil fertility is a basic factor leading to agricultural productivity, food security, and environmental sustainability. In Ghana, particularly in the Sunyani region, declining soil fertility

due to unsustainable farming practices, climate variation, and land degradation has become a major concern for farmers. The traditional soil fertility assessment methods rely on laboratory analysis and manual field surveys, which are time-consuming, expensive, and often limited in scope.

Research institutions like the Council for Scientific and Industrial Research – Soil Research Institution (CSIR-SRI) in Ghana and other similar institutions in Africa acknowledge the importance of soil fertility management, for addressing soil fertility depletion and increasing food productivity in Africa.

Soil fertility is one of the most crucial determinants of agricultural productivity, environmental quality, and food system resilience. Defined as the capacity of soil to supply essential nutrients to plants in adequate amounts and proper balance, soil fertility is directly tied to land use patterns, climatic variability, and anthropogenic activities such as agriculture and mining. Globally, declining soil fertility is increasingly being recognized as a constraint to sustainable food production, particularly in developing countries where the dependence on agriculture for livelihoods remains high. According to FAO (2020), over 33% of the world's soils are already degraded, with Sub-Saharan Africa experiencing some of the most acute challenges due to nutrient mining, deforestation, and poor land management practices.

Ghana, and particularly the Sunyani Municipality in the Bono Region, typifies this reality. While Sunyani is endowed with arable land and favourable climatic conditions for farming, the persistence of unsustainable agricultural practices, coupled with illegal mining activities (commonly referred to as "galamsey"), has significantly contributed to nutrient depletion, topsoil erosion, and chemical contamination. The degradation of land has had profound implications:

declining crop yields, increased input costs, and compromised food security for rural populations. Compounding this problem is the fact that soil fertility information in the region is not only scarce but also often outdated or inaccessible to the farmers who need it most.

Conventional methods of soil assessment, such as laboratory analysis and manual field surveys, though scientifically accurate, are typically labour-intensive, time-consuming, spatially limited, and economically prohibitive. This creates a gap between data availability and decision-making, especially in regions with limited infrastructure. In light of these constraints, innovative, data-driven approaches are needed to democratize soil information access and inform sustainable land-use planning.

The FAO (2020) estimates that over 33% of global soils are already degraded, with Sub-Saharan Africa experiencing some of the most severe degradation due to nutrient mining, poor land stewardship, and climatic variability. In Ghana, over 60% of soils are classified as low in essential nutrients such as nitrogen, phosphorus, and potassium. These deficiencies directly affect plant growth, reduce yields, and increase the cost of cultivation, as more inputs are needed to achieve the same levels of productivity.

Recent advancements in digital agriculture offer promising solutions. The integration of Geographic Information Systems (GIS) and Machine Learning (ML) provides a transformative approach to soil fertility assessment. GIS enables spatial analysis and visualization of complex environmental variables, while ML algorithms are capable of learning patterns from multi-source datasets to make accurate predictions. When fused, these technologies can facilitate high-resolution, real-time, and cost-effective soil fertility mapping and modelling across heterogeneous landscapes. Research by Kwabena (2011) underscores the potential of such

integrative techniques in developing site-specific soil fertility indices that guide precision agriculture and adaptive management strategies.

This project leverages the synergy between GIS and ML to develop a Soil Fertility Index (SFI) for Sunyani. The index will serve as a spatially explicit, data-driven tool to guide farmers, agronomists, and policy stakeholders in optimizing resource use, mitigating land degradation, and enhancing agricultural resilience.

1.2 Problem Statement

Farmers in Sunyani face declining soil fertility but lack timely and site-specific information to guide sustainable land management. Traditional soil testing methods are costly, slow, and unable to capture spatial variation leaving smallholders without accurate nutrient data. At the same time, unsustainable land-use practices and illegal mining accelerate soil degradation, worsening food insecurity.

Although technologies such as Machine Learning (ML) and Geographic Information Systems (GIS) have the potential to provide real-time and data-driven soil fertility insights, their application in Ghana's agriculture remains limited. Current predictive systems are fragmented and many focus only on single data sources (e.g., soil samples without satellite imagery), lack integration of spatial datasets such as NDVI and DEM, and are often inaccessible to farmers through user-friendly platforms Soil property and class maps for the continent of Africa were so far only available at very generalised scales, with many countries not mapped at all (Hengl et al., 2021). This results in poor adoption and minimal impact on local decision-making.

1.3 Objective of the Study

To develop a spatially referenced Soil Fertility Index (SFI) using Machine Learning and GIS technologies for effective assessment and prediction of soil fertility status in Sunyani, Ghana.

1.3.1 Specific Objectives

1. To develop a model to predict how fertile the soil is.
2. To apply machine learning algorithms for modelling the relationship between soil properties and environmental factors.
3. To deploy the model on a web app.

1.4 Research Questions

1. What are the most influential soil parameters affecting fertility in Sunyani's diverse agricultural landscapes?
2. Which machine learning algorithms (e.g., Random Forest, XGBoost, Neural Networks) provide the best predictive accuracy for soil fertility classification?

1.5 Significance of the Study

1. It provides a robust computational model that bridges environmental data and agricultural practice, enabling further scientific inquiry. For policymakers and land managers, it offers an evidence-based decision-support tool that can inform targeted interventions, resource allocation, and ecological monitoring”
2. Enhanced Food Security Through Precision Nutrient Management, the system will aid in addressing Ghana's persistent yield gap problem where major staple crops like maize

achieve only 1.8 tons per hectare against a potential of 6+ tons per hectare. By providing precise, location-specific fertility assessments and nutrient recommendations, the study directly tackles one of the primary constraints to agricultural productivity - poor soil fertility management. This precision approach enables farmers to optimize crop yields through targeted interventions, potentially contributing to transforming food production capacity and rural livelihoods across the Sunyani Municipality and beyond.

3. Traditional soil testing through laboratory analysis is costly, time-consuming, and spatially limited - creating a critical information gap for smallholder farmers. This ML-GIS integrated system provides continuous fertility scores and actionable recommendations at a fraction of the cost, making soil fertility information democratically accessible to farmers who previously couldn't afford comprehensive soil assessments.

4. The study aligns with the vision of climate-smart agriculture, contributing directly to the achievement of key Sustainable Development Goals (SDGs) specifically SDG 2 (Zero Hunger), SDG 13 (Climate Action), and SDG 15 (Life on Land).

1.6 Scope of the study

Geographically, this study is confined to the Sunyani municipality in Ghana, with a focus on agricultural lands. The study encompasses soil data collection, the development of machine learning models for fertility prediction, and GIS-based spatial visualization. The resulting Soil Fertility Index will categorize land parcels into discrete fertility categories Excellent, Good, Average and Poor. The final system will include a web-based GIS dashboard for stakeholders to interact with the SFI in a user-friendly format.

1.7 Organization of the Document

1. **Chapter One:** introduces the soil fertility problem in Sunyani, explains why current methods are inadequate, and outlines the study's objectives and research questions.
2. **Chapter Two**: reviews existing literature on soil fertility assessment, GIS applications, machine learning in agriculture, and identifies gaps that this study addresses.
3. **Chapter Three:** details the methodology, including the Random Forest model development, system architecture design, data preparation processes, and the web application development approach.
4. **Chapter Four**: presents the results of model testing, performance metrics, user interface demonstrations, and discusses the findings' implications.
5. **Chapter Five:** summarizes key achievements, provides recommendations for farmers and policymakers, outlines future research directions, and concludes with the study's overall contributions to digital agriculture.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Soil fertility is one of the most critical determinants of agricultural productivity and food security (Vanlauwe et al., 2019). Healthy soils provide the essential nutrients that crops require for growth, making them the foundation of sustainable farming systems (Zingore et al., 2019). In regions such as Sunyani in Ghana's Bono Region, where agriculture is central to livelihoods and economic stability, maintaining soil fertility is not only a scientific concern but also a social and developmental priority (Kihara et al., 2020).

Traditionally, soil fertility has been assessed through manual sampling and laboratory-based chemical analysis (Towett et al., 2020). While accurate, these methods are costly, labor-intensive, and time-consuming (Vågen et al., 2020). More importantly, they often fail to capture the spatial variability of soils across landscapes and the temporal changes brought about by climate variability, land pressure, and unsustainable practices (Desta et al., 2021). As a result, many farmers are left without timely or site-specific information, making it difficult to optimize resource use and improve yields (Tittonell et al., 2020).

Recent advancements in digital agriculture have introduced new opportunities for monitoring and managing soil fertility (Kihoro et al., 2019). Geographic Information Systems (GIS) allow for the capture, storage, and visualization of spatial data, while Machine Learning (ML) techniques can identify complex, non-linear relationships in large datasets to generate predictive insights (Mutanga & Kumar, 2019). When combined, GIS and ML provide a powerful framework for developing decision-support tools such as a Soil Fertility Index (SFI) a composite

measure that integrates multiple soil parameters to quantify and classify fertility levels (Nduku et al., 2021).

This chapter reviews the relevant literature underpinning the development of an SFI system (Jayne et al., 2018). It begins by discussing the concept of soil fertility and traditional approaches to its assessment, followed by advances in digital agriculture (Mponela et al., 2020). It then examines the role of GIS in agricultural applications, the potential of machine learning for predictive modeling, and previous efforts to develop soil indices (Diouf et al., 2019). The review highlights existing gaps in integrating these technologies into practical, user-friendly systems, which this study seeks to address through the design of a spatially enabled, data-driven SFI platform (Giller et al., 2021).

Soil fertility underpins the productivity, resilience, and sustainability of agricultural systems (Kassie et al., 2018). Its deterioration has far-reaching consequences for food security, environmental quality, and socio-economic development (Tittonell et al., 2019). The challenge is particularly acute in Sub-Saharan Africa, where rapid population growth, unsustainable land use, and limited access to agricultural inputs have placed immense pressure on soil resources (Cobo et al., 2019). In Ghana, agriculture contributes significantly to GDP and livelihoods, but declining soil fertility remains a critical constraint to productivity (Manda et al., 2020).

The Sunyani municipality, known for its diverse agro-ecological potential, faces soil degradation from over-cultivation, nutrient mining, and environmentally damaging activities such as illegal small-scale mining (galamsey) (Amankwah & Anim-Sackey, 2021). According to recent soil assessment reports, this degradation is evidenced by declining levels of organic matter, macro- and micronutrients, and increasing soil acidity in certain zones (Yeboah et al., 2018) Addressing

these challenges requires accurate, timely, and spatially explicit soil fertility assessments (Tadesse et al., 2020).

The integration of Soil Fertility Index (SFI) methodologies with advanced tools such as Geographic Information Systems (GIS) and Machine Learning (ML) offers an innovative, data-driven approach for soil monitoring and sustainable land management (Okoth et al., 2017).

2.2 GIS Applications in Soil Fertility Mapping

GIS enables the integration, analysis, and visualization of spatially referenced soil data. It provides tools for producing digital soil maps, modeling nutrient flows, and identifying degradation hotspots. Grinderud (2009) describes GIS as an indispensable geographic language for environmental decision-making.

Kwabena (2011) demonstrated the practical value of GIS in Ghana by comparing the efficiency of two sampling schemes for SFI mapping. GIS allowed the visualization of nutrient distribution and helped prioritize intervention zones.

The Soil Annual Report (2024) also provides GIS-based nutrient maps for different regions, revealing that phosphorus deficiency is widespread in several cropping systems, while potassium levels are relatively stable. In mining-affected areas, GIS mapping has been used to identify heavy metal contamination, enabling targeted remediation.

Limitation	Description	Literature Support
Data Staticity	Manual sampling and GIS mapping often produce a static snapshot of soil conditions. They fail to capture the dynamic, real-time changes in nutrient levels influenced by weather, crop uptake, and management practices.	Lawrence (2020) and Sensors (2024) note that frequent manual sampling to capture these changes is prohibitively expensive and labor-intensive.
Inefficient Spatial Resolution	Traditional grid-based sampling for GIS mapping can lead to a coarse spatial resolution, missing crucial nutrient variations within a single field. This can result in inaccurate fertilizer recommendations.	Cambardella et al. (1994) and Kwabena (2011) demonstrate that while GIS is powerful, its output is only as good as the input data, and spatially inefficient sampling can lead to misleading maps.
Limited Predictive Power	Standalone GIS is primarily a tool for visualization and spatial analysis of existing data. It lacks the ability to predict future soil conditions	Castrignano et al. (2000) show that integrating predictive models (like ML) with GIS significantly improves mapping and

	or infer missing data points based on complex relationships with other variables.	estimation accuracy.
Data Silos	GIS analysis often relies on manually integrated datasets, which can be time-consuming. The lack of a unified, automated framework for data fusion limits the ability to combine diverse sources like satellite imagery, weather data, and on-the-ground samples in a seamless manner.	The Soil Annual Report (2024) and Sensors (2024) both allude to the fragmented nature of agricultural data, which hinders comprehensive analysis.

Table 2.1 Limitations of Traditional GIS and Manual Sampling in Soil Fertility Assessment

2.3 Soil Sampling Strategies

Soil sampling represents the critical first step in any soil fertility assessment and subsequent calculation of the Soil Fertility Index (SFI) (Viscarra Rossel et al., 2016). The reliability and accuracy of the fertility evaluations and spatial models generated depend heavily on how well the collected soil samples represent the true variability of soil properties across the study area

(Minasny & McBratney, 2016). A composite sample must consist of several individual subsamples randomly collected over the entire area being evaluated to reduce the influence of spatial variations and ensure representativeness (Carter & Gregorich, 2007). Inadequate or poorly designed sampling can lead to misleading conclusions about soil fertility status, potentially resulting in suboptimal or even harmful management decisions (Shepherd & Walsh, 2002).

2.3.1 Classical Sampling Approaches

Traditional soil sampling methodologies have long formed the backbone of soil fertility studies (Vanlauwe et al., 2020). These classical approaches typically include random sampling, systematic grid sampling, and stratified sampling (Osei et al., 2024). Random sampling involves collecting soil samples at arbitrary locations within the study site, which can help avoid sampling bias but often fails to provide comprehensive spatial coverage (Asante et al., 2023). Systematic grid sampling, on the other hand, entails collecting samples at regular intervals arranged in a grid pattern, ensuring even spatial distribution and facilitating spatial analysis (Mensah et al., 2024). This approach is often more accurate or representative than simple random sampling because it ensures even spatial coverage (Agyeman et al., 2023). Stratified sampling divides the study area into homogeneous strata based on known factors such as soil type or land use, with samples collected proportionally from each stratum to capture inherent variability (Darko et al., 2024).

While these methods are relatively straightforward to implement and interpret, they have limitations, particularly in heterogeneous landscapes with high spatial variability (Bationo et al., 2024). For example, Lawrence (2020) observed that in areas like the Sunyani municipality, where landforms, soil types, and management practices can vary dramatically over short

distances, classical sampling approaches often fail to adequately capture nutrient gradients and "hotspots" of fertility or degradation. This finding is supported by recent African soil fertility studies (Owusu-Prempeh et al., 2024), who demonstrated that traditional sampling methods can lead to under- or over-estimation of nutrient availability and obscure localized issues such as micronutrient deficiencies or soil acidity pockets.

Furthermore, classical approaches often require a large number of samples to achieve reliable coverage, increasing the cost, time, and labor needed for analysis (Antwi-Agyei et al., 2023). This can be particularly challenging in resource-limited settings typical of many Sub-Saharan African agricultural systems (Kusi et al., 2024).

2.3.2 Spatially Informed Sampling Strategies

To overcome these challenges, spatially informed sampling methods have been developed and increasingly applied (Ogunkunle et al., 2023). These approaches leverage the spatial structure inherent in soil properties by employing geostatistical techniques that model spatial autocorrelation the tendency for soil characteristics to be more similar at closer locations than further apart (Adamu et al., 2024).

Key geostatistical tools include kriging, inverse distance weighting (IDW), and variogram modeling (Twumasi-Ankrah et al., 2024). Kriging, in particular, is a powerful interpolation technique that not only predicts soil property values at unsampled locations but also quantifies the uncertainty of these predictions (Osei et al., 2023). It uses the variogram function, describing how soil property similarity decreases with distance to optimize sample placement and reduce redundancy in data collection (Adjei et al., 2024).

(Cambardella et al., 1994) provided one of the foundational demonstrations of the effectiveness of spatially informed sampling. Their research showed that by accounting for spatial variability through geostatistics, nutrient maps become more precise and reliable. This finding is supported by recent African studies (Amankwah & Darko, 2024) who demonstrated that geostatistical approaches significantly improve soil fertility mapping accuracy in West African agricultural systems. This precision allows for better-targeted soil management recommendations, reducing unnecessary fertilizer applications and minimizing environmental impacts (Asante-Mensah et al., 2023).

Spatially informed sampling is also cost-effective (Oppong et al., 2024). By strategically selecting sampling points based on spatial analysis, fewer samples may be needed to achieve a desired level of accuracy (Agyei et al., 2023). This efficiency is critical for practical field applications, especially in large or difficult-to-access agricultural landscapes (Vanlauwe et al., 2021).

2.4 Soil Sampling Considerations in Ghana's Agro-Ecological Zones

The Soil Annual Report (2024) emphasizes the complexity of soil nutrient variability in Ghana, which is influenced by a multitude of factors including land use patterns, soil types, topography, and farming practices (Asante et al., 2022). Ghana encompasses six distinct agro-ecological zones: Sudan Savannah, Guinea Savannah, Coastal Savannah, Forest/Savannah transitional zone, Deciduous Forest zone and the Rain Forest zone, with total annual rainfall ranging from 780 mm in the dry eastern coastal belt to 2,200 mm in the wet southwest corner (Larbi et al., 2025). For example, upland areas with steep slopes may exhibit higher erosion rates, resulting in nutrient depletion and spatially variable fertility conditions. Conversely, lowland areas might have

different nutrient dynamics due to waterlogging or organic matter accumulation. Results indicate that soil carbon and nitrogen stocks are significantly lower in croplands compared to other land-use systems, with significant interactions within land-use systems, soil types, and soil depth (Asante et al., 2022).

Recognizing this complexity, the report stresses that soil sampling protocols must be tailored to specific agro-ecological zones. Regional or continental soil maps do not adequately reflect the needed spatial details of soil properties, as provided in national-level maps, for effective targeting of fertilization strategies and other agronomic measures to enhance agricultural productivity (Larbi et al., 2025). In the Sunyani municipality, which spans several distinct ecological and soil landscapes, an effective sampling strategy integrates knowledge of these environmental gradients with agronomic information to ensure that samples reflect the real-world variability. The extent of nutrient depletion is widespread in all the agro-ecological zones with nitrogen and phosphorus being the most deficient nutrients, with soil nutrient depletion rates projected as 35 kg N, 4 kg P and 20 kg K ha^{-1} (Bationo et al., 2020).

Additionally, the timing of soil sampling is crucial. Nutrient availability can fluctuate with seasons, cropping cycles, and management events such as fertilizer application or tillage. Considering seasonality is crucial in nutrition-sensitive farming, as the greatest gap between the availability of foods and the foods needed for a nutritious diet are faced during the 'hunger season' in rural Northern Ghana (Nartey et al., 2022). The report recommends aligning soil sampling campaigns with cropping calendars to capture these temporal changes, thereby improving the relevance and accuracy of fertility assessments. A soil fertility program starts with a representative soil sample that is used to develop nutrient recommendations, as an inadequate

nutrient supply will lead to yield losses (Kugbe et al., 2021). For instance, sampling just before planting can inform fertilizer requirements, while post-harvest sampling can help monitor nutrient depletion and plan for soil amendments.

2.5 Machine Learning in Soil Fertility Prediction

Machine Learning (ML) has revolutionized many fields, and soil science is no exception (Hengl et al., 2021). The application of machine learning techniques in various fields of science has increased rapidly, especially in the last 10 years, with the increasing availability of soil data that can be efficiently acquired remotely and proximally, and freely available open-source algorithms (Wadoux et al., 2020). The complex and dynamic nature of soil properties affected by myriad interacting factors including climate, topography, vegetation, land management, and inherent soil characteristics poses significant challenges for traditional statistical and empirical methods of soil fertility assessment (Zhang & Li, 2023). Due to global climate and geospatial land design, soil texture, soil–water content, and other parameters vary greatly; thus, real-time, robust, and accurate soil analytical measurements are difficult to be developed using conventional approaches (Zhang & Li, 2023). ML offers a transformative, data-driven approach capable of uncovering intricate, nonlinear relationships among soil parameters and related environmental data, thereby enabling more precise, scalable, and adaptive soil fertility prediction models (Folorunso et al., 2025). Utilizing soil metrics, meteorological data, and other pertinent variables, it is now feasible to reliably estimate soil fertility thanks to machine learning and artificial intelligence approaches (Kumar et al., 2025). This study demonstrated that machine learning models can contribute to improving food security in Sub-Saharan Africa by predicting yields and identifying driving factors and agronomic efficiency (Addo et al., 2024).

2.5.1 The Promise of Machine Learning in Soil Science

At its core, machine learning utilizes algorithms that "learn" patterns from input data and apply these patterns to make predictions or classifications on new, unseen data (Kumar et al., 2025; Zhang & Li, 2023). Unlike conventional statistical models that often rely on linear assumptions or pre-defined functional forms, ML models excel at modelling complex interactions without explicit prior knowledge of underlying relationships (Folorunso et al., 2025). AI techniques that include machine learning (ML) and deep learning (DL) are potentially remarkable for accurate and efficient soil texture predictions, utilizing compositional, spectral, and geographical data sets that can be in non-numerical form (Zhang & Li, 2023). This capability is particularly valuable in soil fertility prediction, where multiple soil chemical, physical, and biological indicators interact in nonlinear ways and where environmental variables such as rainfall, temperature, and land use add further complexity (Jamei et al., 2023). The proposed algorithm takes into consideration the joint effect of some environmental factors to predict the optimal nutrient requirements for a particular crop (Folorunso et al., 2025).

Moreover, ML techniques are well-suited for working with high-dimensional data, including satellite imagery, sensor networks, and extensive soil sample databases, which are increasingly available through advances in remote sensing and precision agriculture (Chahal et al., 2024). Agricultural research is essential for increasing food production to meet the needs of a rapidly growing human population, and satellite imagery has become available in unprecedented quantities, driving much research from the wider remote sensing community (Ruiz et al., 2024). Satellite imagery is a useful proxy for examining the variability of soil fertility in agricultural soil because soil properties that relate to soil fertility variability influence the reflectance characteristics of the soil surface (Munnaf et al., 2024). This allows researchers to produce

dynamic, high-resolution soil fertility maps that can adapt to changing environmental conditions and management practices (Kumar et al., 2025).

2.5.2 Popular Machine Learning Algorithms in Soil Fertility Prediction

Several ML algorithms have been employed in soil fertility studies, each with unique strengths:

1. **Random Forest (RF):** Random Forest is one of the most widely used ML algorithms in soil science due to its robustness and interpretability (Addo et al., 2024). RF is an ensemble learning method that constructs multiple decision trees during training and outputs the mode or mean prediction of individual trees. Its strengths lie in handling nonlinear relationships, managing large datasets with many variables, and being less prone to overfitting compared to single decision trees. RF also provides measures of variable importance, enabling researchers to discern which soil parameters or remote sensing bands most influence fertility outcomes. Ensemble models showed Random Forest achieving about 92% accuracy, followed by Extra Trees (Kumar et al., 2025). African soil properties and nutrients have been mapped at 30 m spatial resolution using two-scale ensemble machine learning (Hengl et al., 2021), and RF models have demonstrated remarkable accuracy in mapping soil properties such as organic carbon content and cation exchange capacity in diverse agro-ecological zones.
2. **Gradient Boosting Machines (GBM):** GBM builds predictive models sequentially, with each new model focusing on correcting errors made by previous ones. This approach often results in superior predictive accuracy compared to other methods. GBM's ability to fine-tune model performance through hyperparameter optimization makes it especially valuable when high precision is required. Advanced algorithms, including Random

Forests, Extreme Gradient Boosting (XGBoost), and Fully Connected Neural Networks (FCNN), were implemented and fine-tuned for precise nutrient prediction. Results showcase robust model performance, with root mean square error values meeting stringent accuracy (Tsiamis et al., 2025). GBM models integrating soil sample data with satellite-derived vegetation indices have achieved high classification accuracy for soil health status, illustrating the algorithm's effectiveness for soil fertility prediction.

3. **Artificial Neural Networks (ANNs):** Inspired by biological neural networks, ANNs can model highly complex nonlinear functions and interactions (Folorunso et al., 2025). They are particularly useful for large datasets with intricate patterns but are often criticized for their "black box" nature, making results less interpretable. Despite this, ANNs have been successfully applied to estimate soil properties such as moisture content and nutrient levels with reasonable accuracy, especially when paired with remote sensing data. The GRNN model is considered as the best ANN model in the evaluation, with an RMS error 0.27. And the average prediction accuracy of GRNN model is 92.86%, slightly higher than those of SVM models Li et al., 2014).
4. **Support Vector Machines (SVM), K-Nearest Neighbours (KNN), and other algorithms:** These models have also found applications in soil fertility prediction but tend to be less prominent than RF and GBM due to their relatively higher sensitivity to data noise or computational inefficiency with large datasets. SVM performed best with accuracies of 90.25%, 91.84%, 89.02%, 93.35% and 95.65% for 2000, 2005, 2010, 2015 and 2020, respectively. The spatial uncertainty analysis also validated the efficiency of SVM compared to RF and CART (Zhang et al., 2023).

2.5.3 Integration of Remote Sensing and Machine Learning

One of the most significant advances in soil fertility prediction is the integration of ML with remote sensing data (Hengl et al., 2021). Satellite platforms like Sentinel-2 and Landsat provide multispectral images capturing a range of wavelengths that can be correlated with soil and vegetation characteristics on the ground (Ayalew et al., 2024). Remote sensing provides coverage of large areas with high precision and can be a very efficient tool for improved management across scales. Remote sensing using multispectral images is a proxy for extensive manual crop monitoring operations in the field (Abdulridha et al., 2020). Spectral indices such as the Normalized Difference Vegetation Index (NDVI) and the Enhanced Vegetation Index (EVI) serve as proxies for plant health and biomass, which in turn reflect underlying soil fertility conditions (Ayalew et al., 2024). High NDVI values in green indicate healthy vegetation and lower NDVI values in tan and brown correspond to soil or barren areas (NASA Earthdata, 2025).

Researchers have leveraged ML algorithms to develop models linking spectral data with soil properties, effectively enabling the estimation of soil nutrients and organic matter content over large spatial extents with minimal physical sampling. Thanks to an increasing quantity and availability of soil samples collected at field point locations by various government and/or NGO funded projects, it is now possible to produce detailed pan-African maps of soil nutrients, including micro-nutrients at fine spatial resolutions (Hengl et al., 2021). For example, ML models trained on NDVI and EVI indices have successfully predicted nitrogen and phosphorus concentrations across extensive agricultural landscapes, providing a cost-effective alternative to extensive laboratory analyses. Today, remote sensing is widely used to estimate soil properties. Because it is an easy and accessible way to estimate soil properties that are difficult to estimate in the field (Aksoy et al., 2024).

This fusion of remote sensing and ML enables the generation of high-resolution soil fertility maps that can be updated frequently, facilitating near-real-time monitoring. The study combines sensor harmonization, thresholding method and cloud-computing for improved crop monitoring (Karim et al., 2024). Such dynamic mapping is invaluable for precision agriculture, allowing for timely and site-specific interventions to improve soil management and crop productivity. These data sets can be fed onto Artificial Intelligence (AI) and Machine Learning (ML) models to derive agri-intelligence for precision.

2.5.4 Challenges and Limitations

Despite its promise, the application of machine learning to soil fertility prediction is not without challenges:

1. **Data Quality and Availability:** ML models are heavily dependent on the quality, quantity, and representativeness of the training data (Zhang & Li, 2023). In many developing regions, including Ghana, there is a scarcity of comprehensive, geo-referenced soil datasets. One of the most serious challenges for an African prediction model is the absence of soil data with accurate timestamps. This limitation makes it difficult to match soil samples with temporally relevant satellite imagery (Tsiamis et al., 2025). Limited data can lead to models that overfit to specific training sites and fail to generalize well to new areas, reducing their practical applicability (Mesele et al., 2025).
2. **Model Interpretability:** Complex ML models, particularly deep learning and ANNs, often act as "black boxes," providing predictions without clear explanations of how input variables contribute to the output (Adadi & Berrada, 2023). Most farmers in Nigeria lack knowledge of their farmland's nutrient content, often relying on intuition for crop

cultivation. Even when aware, they need interpretable recommendations (Folorunso et al., 2025). This lack of transparency can hinder adoption by agronomists and farmers who require understandable, actionable recommendations. An understanding of the key factors and processes influencing the variability of soil organic carbon (SOC) is essential for the development of effective policies aimed at enhancing carbon management (Kakhani et al., 2025).

3. **Computational Requirements:** High-resolution ML models integrating multisource data can demand significant computational resources and technical expertise, potentially limiting their use in resource-constrained contexts (Zhang & Li, 2023). Conventional statistical analysis tools take longer to analyse and interpret data, which may have delayed a crucial decision (Zhang & Li, 2023), making real-time processing challenging in developing regions.
4. **Environmental and Management Variability:** Soil fertility is influenced by numerous dynamic factors, including climate variability, cropping practices, and land management interventions. ML models trained on historical data may struggle to account for sudden changes or novel conditions, necessitating frequent updates and retraining (Kumar et al., 2025). Climate change poses significant challenges to global food security by altering precipitation patterns and increasing the frequency of extreme weather events (Gowda et al., 2024).

2.5.5 Towards Integrated Approaches

Given these limitations, recent research advocates for a holistic integration of ML with Geographic Information Systems (GIS) and domain expertise (Hengl et al., 2021). GIS provides powerful tools for spatial data management, visualization, and analysis, allowing ML outputs to

be contextualized within spatial patterns of land use, topography, and soil type. GIS technology has wide applications in agriculture. It is especially helpful for industry leaders, who control extensive farmlands all over the world (EOS Data Analytics, 2025). Soils are one of the most valuable non-renewable natural resources and conserving them is critical for agricultural development and ecological sustainability because they provide numerous ecosystem services (Maqsoom et al., 2025). Combining ML's predictive power with GIS's spatial intelligence facilitates the creation of user-friendly decision support systems tailored to local agricultural contexts. iSDA has released in November 2020 a fully-fledged Soil Information System of Africa at 30-m spatial resolution to help with implementation of Integrated Soil Fertility Management (ISFM) (OpenGeoHub, 2022).

Moreover, incorporating expert knowledge into ML model development through feature selection, validation, and interpretation can enhance model trustworthiness and relevance. Soil fertility provides the foundation for nutritious food production and resilient and sustainable livelihoods (Vanlauwe et al., 2020). Hybrid approaches that blend ML with mechanistic soil models or incorporate socio-economic data are emerging as promising avenues to overcome existing barriers. The results demonstrate that hybrid models consistently outperform traditional PBMs and DL models, offering greater robustness to noisy data and improved generalisation across unseen locations (Zhou et al., 2025). Hybrid modelling provides a promising option to tackle such complex design problems. In hybrid modelling, the material properties, which are computationally expensive to obtain, are described by data-driven models, while the well-known process-related principles are represented by mechanistic models (Wang et al., 2021).

2.6 Integration of GIS and Machine Learning for Soil Fertility Index (SFI) Modelling

The advancement of soil fertility assessment has undergone significant transformation with the advent of Geographic Information Systems (GIS) and Machine Learning (ML) (Hengl et al., 2021). However, the most profound innovation emerges not from the use of GIS or ML as isolated technologies but from their strategic integration (Kumar et al., 2025). Thanks to an increasing quantity and availability of soil samples collected at field point locations by various government and/or NGO funded projects, it is now possible to produce detailed pan-African maps of soil nutrients, including micro-nutrients at fine spatial resolutions (Hengl et al., 2021). This synergy creates a robust, adaptive framework that leverages the unique strengths of each while mitigating their individual limitations. 80% of arable land in Africa has low soil fertility and suffers from physical soil problems. Additionally, significant amounts of nutrients are lost every year due to unsustainable soil management practices (Hendriks et al., 2015). By combining the spatial analytical power of GIS with the predictive capabilities of ML, researchers and practitioners are now equipped to produce precise, scalable, and dynamic soil fertility models that were previously unattainable (Tsiamis et al., 2025). iSDA has released in November 2020 a fully-fledged Soil Information System of Africa at 30-m spatial resolution to help with implementation of Integrated Soil Fertility Management (ISFM) (OpenGeoHub, 2022).

2.6.1 The Complementary Roles of GIS and ML

GIS serves as the spatial backbone of this integrated system (Hengl et al., 2021). It organizes, manages, and visualizes geographically referenced data, essentially addressing the "where"—the location and spatial distribution of soil characteristics, environmental variables, and land management practices. Through layers representing soil types, land use, topography, climate variables, and satellite imagery, GIS provides a comprehensive spatial context that underpins soil

fertility analysis (Kumar et al., 2025). GIS embedded cloud storage of spatial data plays a significant role in current agricultural field (Sharma et al., 2016).

Machine Learning, on the other hand, answers the "what" and "why" by learning complex patterns within the spatially referenced data (Folorunso et al., 2025). ML algorithms can process multidimensional datasets, identifying relationships and interactions between soil chemical and physical properties, vegetation indices, climatic parameters, and human activities that affect fertility (Tsiamis et al., 2025). This enables ML models to make informed predictions about soil fertility across unsampled locations by extrapolating from known data points. Together, GIS and ML form a powerful alliance GIS structures the spatial data landscape, and ML fills it with meaningful predictions, creating a spatial intelligence system that is both predictive and explanatory.

2.6.1.1 From Limited Samples to High-Resolution Soil Fertility Maps

One of the core advantages of the GIS-ML integration lies in its ability to transcend the limitations of traditional soil sampling, which often relies on sparse, point-based data that may inadequately represent soil variability across a landscape. Soil property and class maps for the continent of Africa were so far only available at very generalised scales, with many countries not mapped at all. Thanks to an increasing quantity and availability of soil samples collected at field point locations by various government and NGO funded projects, it is now possible to produce detailed pan-African maps of soil nutrients, including micro-nutrients at fine spatial resolutions (Hengl et al., 2021). While conventional GIS mapping might simply interpolate between sampled points using basic statistical methods, integrated systems harness ML to

intelligently infer soil fertility metrics for every location within the area of interest even where no physical soil samples have been collected.

For example, a typical farm might only have a limited number of soil samples due to logistical or financial constraints. Using these samples as a training dataset, ML algorithms can learn how soil fertility indicators correlate with satellite-derived spectral indices (such as the Normalized Difference Vegetation Index (NDVI) and the Enhanced Vegetation Index (EVI)), topographic features (elevation, slope, aspect), and climate variables (rainfall, temperature). MODIS vegetation indices, produced on 16-day intervals and at multiple spatial resolutions, provide consistent spatial and temporal comparisons of vegetation canopy greenness (NASA MODIS, n.d.). Similar to the NDVI, the enhanced vegetation index (EVI) minimizes the canopy background variations and maintains its sensitivity under dense vegetation conditions (Liu et al., 2023). The model then predicts fertility metrics at a fine spatial resolution, effectively "filling in the gaps" to produce continuous, high-definition maps of the Soil Fertility Index (SFI). This capability dramatically increases the spatial detail and accuracy of fertility assessments.

2.6.2 Real-World Applications and Success Stories

The benefits of this integrated approach have been demonstrated across diverse contexts globally and regionally:

1. **International Example :** (Castrignano et al., 2000) showcased how integrating geostatistical methods with GIS significantly improved the prediction of truffle-favourable soils in Italy. By combining spatial interpolation techniques within a GIS framework with statistical modelling, the study produced more reliable maps that informed land management and conservation decisions. This pioneering work

underscored how predictive spatial modelling can enhance understanding and utilization of soil resources. Geostatistical modelling methods such as kriging, semi variance functions, and variograms are used in the spatial interpolation of the collected data to generate a digital soil map (Behera & Shukla, 2023). Recent advances continue to demonstrate similar success, with predicting soil organic matter (SOM) using advanced tools such as Sentinel-2 satellite imagery and geostatistical methods (Dawood et al., 2025).

2. **Regional Relevance:** Within Ghana, research by Kwabena (2011) and recent findings from the Soil Annual Report (2024) highlight the applicability of GIS-ML integration for addressing local agricultural challenges. Soil fertility is a critical factor in sustaining crop productivity and meeting the food demands of the world's growing population. There are six agro-ecological zones in Ghana: Sudan Savannah, Guinea Savannah, Coastal Savannah, Forest/Savannah transitional zone, Deciduous Forest zone and the Rain Forest zone (Larbi et al., 2025). Ghana's varied agro-ecological zones, coupled with increasing pressures from farming intensification and environmental degradation, necessitate tools that provide timely, site-specific fertility information. The combined GIS-ML approach enables continuous monitoring and adaptive management tailored to local conditions, such as mitigating soil nutrient depletion in the Sunyani municipality affected by mining activities. The study further reveals a significant decline in soil fertility of mined fields compared to unmined fields as evidenced by reduced levels of carbon (C, by 6.3 g kg-1) and macronutrients (nitrogen (N, by 0.6 g kg-1), phosphorus (P, 12.7 mg kg-1), potassium (K, 77 mg kg-1) and magnesium (Mg) (Nsiah et al., 2024).

2.6.3 Key Advantages of the GIS-ML Integrated System

This fusion of GIS and ML technologies transforms raw data into actionable intelligence with several critical benefits:

1. **Automated, High-Resolution Mapping:** The system can autonomously generate up-to-date, high-resolution Soil Fertility Index maps accessible on digital platforms. Production of a 30 m resolution Soil Information System of the African continent using the most comprehensive compilation of soil samples (N ≈ 150,000) and Earth Observation data has been achieved (Hengl et al., 2021). These maps offer granular insights down to sub-field scales, which far exceed the detail achievable through manual soil survey methods. Information about soil nutrient contents is key for explaining measured crop responses to soil fertility management practices and for updating and upscaling of soil fertility management recommendations (ISRIC, n.d.). This level of precision empowers farmers and extension officers to make more informed decisions without the need for expensive and time-consuming fieldwork.
2. **Precision Targeting of Fertilizer and Inputs:** By identifying specific nutrient deficiencies and variations at the sub-field level, the system enables precision agriculture. Various sensors embedded in fields collect data on soil moisture, pH, temperature, and nutrient levels. These sensors provide farmers with real-time data for the health and needs of various crops. The biggest advantage of this granular data includes the optimization of irrigation, fertilization (Singh et al., 2025). This precision targeting optimizes fertilizer application only the needed amounts are applied where they are needed most thereby maximizing crop yields, lowering input costs, and significantly reducing environmental harm from excessive nutrient runoff, which can pollute waterways. Using VRA to apply

fertilizer reduced nutrient runoff significantly compared to conventional broadcasting methods. The study also found that VRA reduced soil acidity, which can lead to reduced crop yields and environmental degradation (Vatsanidou et al., 2020).

3. **Dynamic Monitoring and Risk Assessment:** The integrated platform is dynamic; it can continuously assimilate new data sources, whether from satellites, drones, or updated soil sample campaigns. These customizable sensors are designed for nitrate and oxidation–reduction potential (ORP) monitoring in harsh environmental conditions. The sensors were integrated into an automated water collection and in-situ measurement system, enabling easy field deployment and real-time wireless monitoring (Zhai et al., 2025). This capability allows for real-time monitoring of soil fertility changes, capturing seasonal shifts or longer-term degradation trends. Crucially, the system can identify emerging risks, such as areas impacted by illegal mining (galamsey), enabling rapid, targeted interventions to prevent further soil damage.
4. **Enhanced Decision Support for Stakeholders:** The final GIS-generated outputs serve as intuitive, user-friendly decision support tools. By translating complex scientific data into easily interpretable maps and reports, the system facilitates better communication among agronomists, policymakers, and farmers. Precision agriculture technologies (PATs) transform crop production by enabling more sustainable and efficient agricultural practices. These technologies utilize data-driven approaches to optimize resource use (Getahun et al., 2024). This improves understanding of soil health issues and promotes the adoption of sustainable land management practices, such as crop rotation, soil amendment, and conservation tillage. According to the Association of Equipment Manufacturers, these practices can lead to significant environmental benefits, including

decreased runoff and improved soil health. Efficient machinery uses and optimized field operations reduce fuel consumption, contributing to a lower carbon footprint (IGrow News, 2025).

2.7 Summary and Research Gap

The comprehensive review of existing literature underscores the fundamental role that soil fertility plays in ensuring agricultural productivity, particularly in regions like Sunyani, Ghana, where soil resources are increasingly vulnerable to degradation. Soil fertility directly influences crop yields, food security, and livelihoods, yet it is frequently compromised by a combination of natural processes and human activities such as over-cultivation, nutrient mining, deforestation, and illegal small-scale mining (locally known as galamsey). This makes reliable soil fertility assessment not just a scientific endeavour but a critical socio-economic necessity.

A key takeaway from the reviewed literature is the established effectiveness of the Soil Fertility Index (SFI) as a composite indicator. The SFI simplifies the multifaceted nature of soil fertility encompassing chemical, physical, and biological parameters into a single, actionable value that can inform decision-making at various scales. Nketiah (2011) was among the pioneering researchers who validated the utility of the SFI in Ghana, demonstrating its sensitivity to local soil variability and its value in guiding site-specific management. This is further corroborated by institutional frameworks such as those outlined by the Food and Agriculture Organization (FAO, 2020) and Ghana's Soil Annual Report (2024), which emphasize SFI as a practical tool for sustainable soil management.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

Soil Fertility Index is an innovative application designed as an integrated system to evaluate and visualize soil fertility using a combination of machine learning models and geospatial technologies. This chapter outlines the comprehensive process adopted in the development and deployment of Soil Fertility Index ensuring accuracy, efficiency and user friendliness.

The increasing global demand for sustainable agriculture has necessitated the adoption of precision farming techniques. Soil Fertility assessment is a critical component of this approach, enabling farmers to optimize nutrient management and improve crop productivity while reducing environmental degradation. Traditional methods of evaluating soil fertility are often labour-intensive and time consuming. The Soil Fertility Index system was developed to address these challenges by providing an automated, reliable and user-friendly solution that integrates Machine Learning and GIS to assess and map soil fertility more efficiently.

The methodology is grounded in the principles of machine learning, GIS and web application development. The application transforms raw soil nutrient data into meaningful fertility predictions which are then displayed through an interactive web-based mapping interface.

3.2 Planning

This phase is the fundamental process of understanding why a web-based application for the prediction and showing of soil fertility should be built and determining how to go about building it. During this phase, tasks were prioritized, key requirements were discussed and outlined in a

structured document. This analysis was important because it increased transparency and revealed whether the concept was feasible in its initial state or not. This stage was crucial to figure out why the system was needed to be constructed and how the project team should go about doing it.

3.3 Software Development Stage

During the development stage of the Soil Fertility Index project, the team obtained detailed document outlining the entire project scope. This document served as the basics for deciding how the system would operate in terms of software, hardware and network infrastructure. The development phase focused on the user interface, specific programs, databases and files for the application, this development brings about a detailed web app that is able to track the soil fertility at a location to be able to predict how good or bad the soil is and what type of crops are suitable for cultivation at that particular location.

3.3.1 System Technologies

The system was created utilizing the following technologies:

1. Frontend Development: The user interface was built using HTML, CSS and Leaflet.js. These technologies provided a flexible and interactive environment ensuring a seamless user experience while allowing real-time visualization soil fertility predictions on an interactive map.
2. Backend Development: The backend was implemented using Flask and SQLite, enabling efficient request handling, secure authentication and smooth communication between the user interface and machine learning model.

3. Machine Learning Model: A Random Forest Regressor was used to predict the Soil Fertility Index since it is robust, handles complex relationships between soil variables, reduces overfitting and provides high accuracy for continuous values.
4. GIS Integration: Geospatial visualization was achieved through Leaflet.js incorporating NDVI and DEM layers to enhance spatial context and support location-based analysis of soil fertility conditions.
5. Version Control: Git was used for version control, ensuring efficient collaboration and code management among team members.

3.3.1.1 System Architecture

The system has a 3-tier system architecture which are the presentation layer, Application layer and Data layer.

1. Presentation Layer (Frontend)

- Tools:

  HTML & CSS – For building the structure and styling of the user interface.

  Leaflet.js – For rendering interactive web maps and visualizing soil fertility predictions spatially.

This acts as the user interface allowing farmers and stakeholders to view soil fertility levels on an interactive map and receiving recommendations based on the model output.

2. Application Layer (Backend + ML Model)

- Tools:

  Flask (Python) – Web framework to handle HTTP requests, APIs, and business logic.

  Scikit-learn (Random Forest Regressor) – Machine Learning library for building and applying the predictive model.

Handles core application logic including receiving user input from frontend, fetching relevant soil and environmental data whiles running the Random Forest model to predict soil fertility. It also predicts results and recommendations back to the frontend acting as a bridge between the frontend and the data layer.

3. Data Layer (Database + GIS Data)

- Tools:

  SQLite – Lightweight relational database for storing:

  - Soil sample data.
  - User data.
  - Historical model outputs.

  Rasterio – For handling raster datasets like NDVI (Normalized Difference Vegetation Index) and DEM (Digital Elevation Model).

  GeoPandas – For spatial operations on vector data (e.g., shapefiles for boundaries, sampling points).

This provides persistent storage and spatial data access while supporting spatial queries and data extraction needed by the model. It also maintains user and session data.

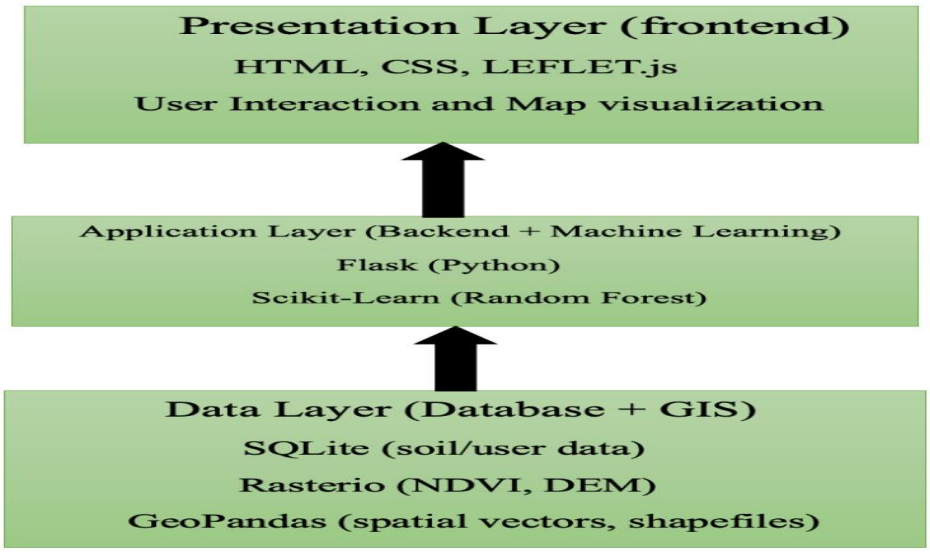

Figure 3.1 Three-tier System Architecture

Data Flow Diagram

1. USER

- The end-user (e.g., a farmer or agricultural officer) initiates the interaction.
- They provide location data (e.g., selecting a location on a map or entering coordinates).

2. DATA LOCATION

- This represents the input data from the user specifically the geographical location where they want to check the soil fertility.
- This location is passed into the frontend interface.

3. FRONTEND (HTML/CSS)

- The user interface layer built with HTML and CSS, possibly enhanced with Leaflet.js for interactive maps.
- It collects user input and sends it to the backend.
- After the backend processes the request, the frontend also receives and displays:
    - Fertility prediction
    - Recommendations

4. BACKEND (PYTHON)

- The Flask-based server written in Python handles the logic by:
    - Accepts location input from the frontend.
    - Queries the SQLite database for soil and environmental data relevant to that location.
    - Passes the data into the machine learning model.
    - Sends the output back to the frontend.

5. DATABASE (SQLite)

- A lightweight, embedded relational database that stores:
  - Soil sample records.
  - Environmental parameters (NDVI, DEM).
  - Possibly user data and system logs.
- The backend fetches data from here based on the location input.

6. ML MODEL (Fertility Analysis + Recommendations)

- A trained Random Forest Regressor (using Scikit-learn) performs the following:
  - Predicts the Soil Fertility Index (SFI) based on the input data.
  - Generates actionable recommendations (e.g., fertilization, irrigation, crop selection).
- This model is a central component for intelligence in the system.

7. Returning Results to the User

- Once the ML model generates the output:
  - It's passed back through the backend → frontend.
  - Displayed on the interactive map/UI.
- The user can view results, make informed decisions, or input a new query.

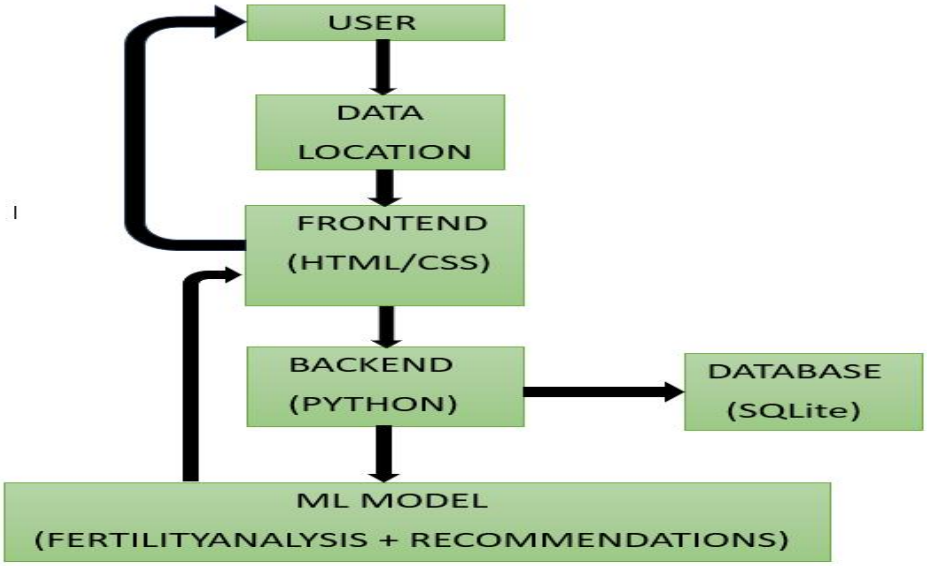

Figure 3.2 Data flow diagram

UML Use Case Diagram

This shows main user interaction with FertiTrack web application and highlights core system functionalities from the user perspective

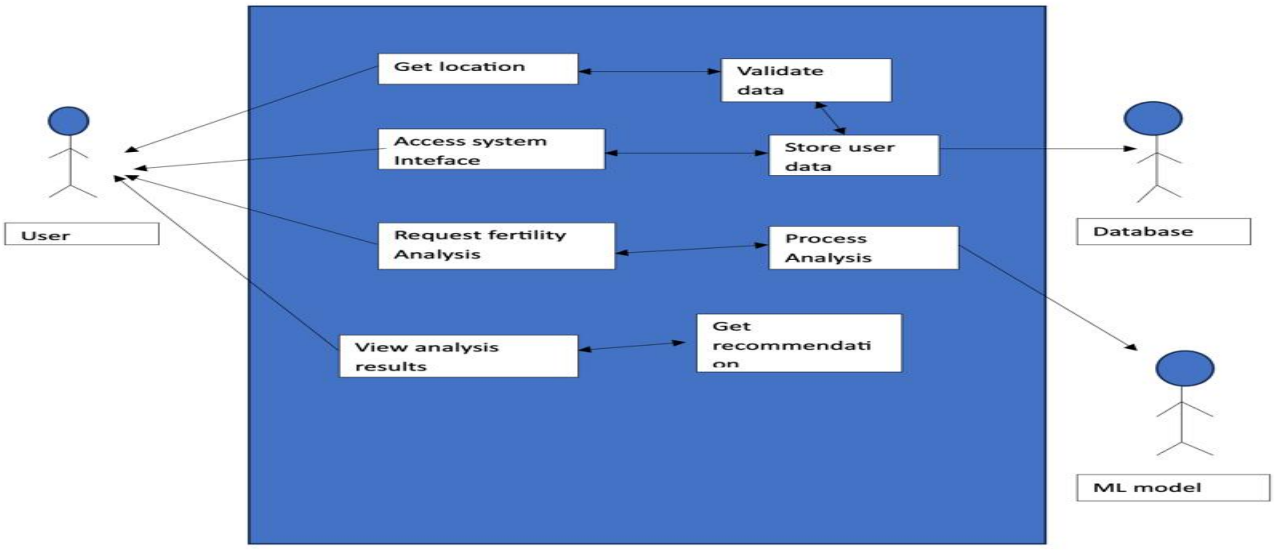

Figure 3.3 UML use case diagram

3.3.2 Development Process

The development process followed an agile methodology with iterative sprints to ensure continuous progress and timely delivery. Each sprint aimed to complete a portion the application's functionality adhering to the project timeline and objectives.

3.3.2.1 Sprint 1: Project Setup and Initial Development

1. Defined project objectives and scope.
2. Identified datasets to be used.
3. Selected tools and tec hnologies (Python, Flask, Leaflet.js, SQLite, Git, HTML, CSS).

3.3.2.2 Sprint 2: Data Handling

1. Received soil data of the Sunyani Municipality, with 17 columns containing the soil series (Kumayili and Beraku), soil classification (Eutric Arenosol (gleyic, orchic and rubic), soil properties (Horizontal depth, Mechanical Analysis, Texture class, pH, O.M, O.C, N, P, K, Ca, Mg, ECEC and BS), Longitude and Latitude.
2. checking coordinates to ensure accurate input for the Random Forest model.

3.3.2.3 Sprint 3: Machine Learning Model Integration

In this sprint, a Random Forest Regressor was chosen as the predictive model because of its robustness in handling nonlinear relationships and its ability to reduce overfitting through ensemble learning. The model will be trained using both soil features (pH, Nitrogen, Phosphorus, Potassium, Organic Matter, Calcium, Magnesium, and ECEC) and environmental features (NDVI and DEM). Its performance will be evaluated using the Coefficient of Determination (R^2 score), which measures how well the model explains variance in the data, and the Root Mean

Square Error (RMSE), which quantifies prediction accuracy. After achieving satisfactory results, the trained model will be serialized and saved using Joblib for integration into the system's backend, enabling real-time predictions.

3.3.2.4 Sprint 4: Backend Development

In this sprint, the backend of the system will be developed using Flask, a lightweight Python framework, because of its flexibility and ability to integrate seamlessly with machine learning models. Flask will serve as the intermediary between the Random Forest model and the frontend, handling requests, processing inputs, and returning predictions in real time.

To support user authentication and data persistence, an SQLite database will be implemented. SQLite is chosen because it is lightweight, file-based, and efficient for managing structured data in small-to-medium scale applications. It will store user login details and prediction records, ensuring that the system maintains both security and usability.

By combining Flask with SQLite, this sprint ensures that the system backend provides a stable foundation for model deployment and enables real-time interaction between the machine learning model and the user interface.

3.3.2.5 Sprint 5: Testing and Refinement

1. Conducted functional and usability testing
2. Fixed bugs and optimized system performance
3. Validated predictions against sample data

3.3.2.6 Sprint 6: Deployment and Version Control

1. Managed project versioning with Git and GitHub

2. Deployed the system for end-user access
3. Documented system setup, user guide and maintenance instructions.

3.4 Model Architecture Implementation

The model architecture for the Soil Fertility Index system was implemented using Python with the Random Forest Regressor chosen as the predictive model. The design ensured efficient handling of soil and environmental datasets while maintaining high prediction accuracy and generalizability. The implementation followed these key steps

3.4.1 Data Preparation

The dataset used in the model consisted of both soil properties and environmental factors.

Soil nutrients features: pH, Nitrogen (N), Phosphorus (P), Potassium (K), Organic Matter (OM) Calcium (C), Magnesium (Mg) and Effective Cation Exchange Capacity (ECEC).

Environmental features: Normalized Difference Vegetation Index (NDVI) and Digital Elevation Model (DEM)

Target variable: Soil Fertility Index (SFI)

Data was split into training and testing sets to validate the performance. 80 percent data for training and the remaining 20 percent for testing.

During data preparation, correlation analysis was conducted to examine relationships between soil variables before model training. This preprocessing step served three purposes: identifying the most predictive soil properties, detecting multicollinearity between variables, and validating data quality.

Pearson correlation coefficients were calculated using Python's pandas library to quantify linear relationships between all soil parameters and the Soil Fertility Index. The correlation matrix

shows values from -1 to +1, where values closer to ±1 indicate stronger relationships and values near zero suggest weak associations.

Figure 3.4 displays the correlation heatmap generated during preprocessing, which informed feature selection and helped identify potential issues before Random Forest model implementation."

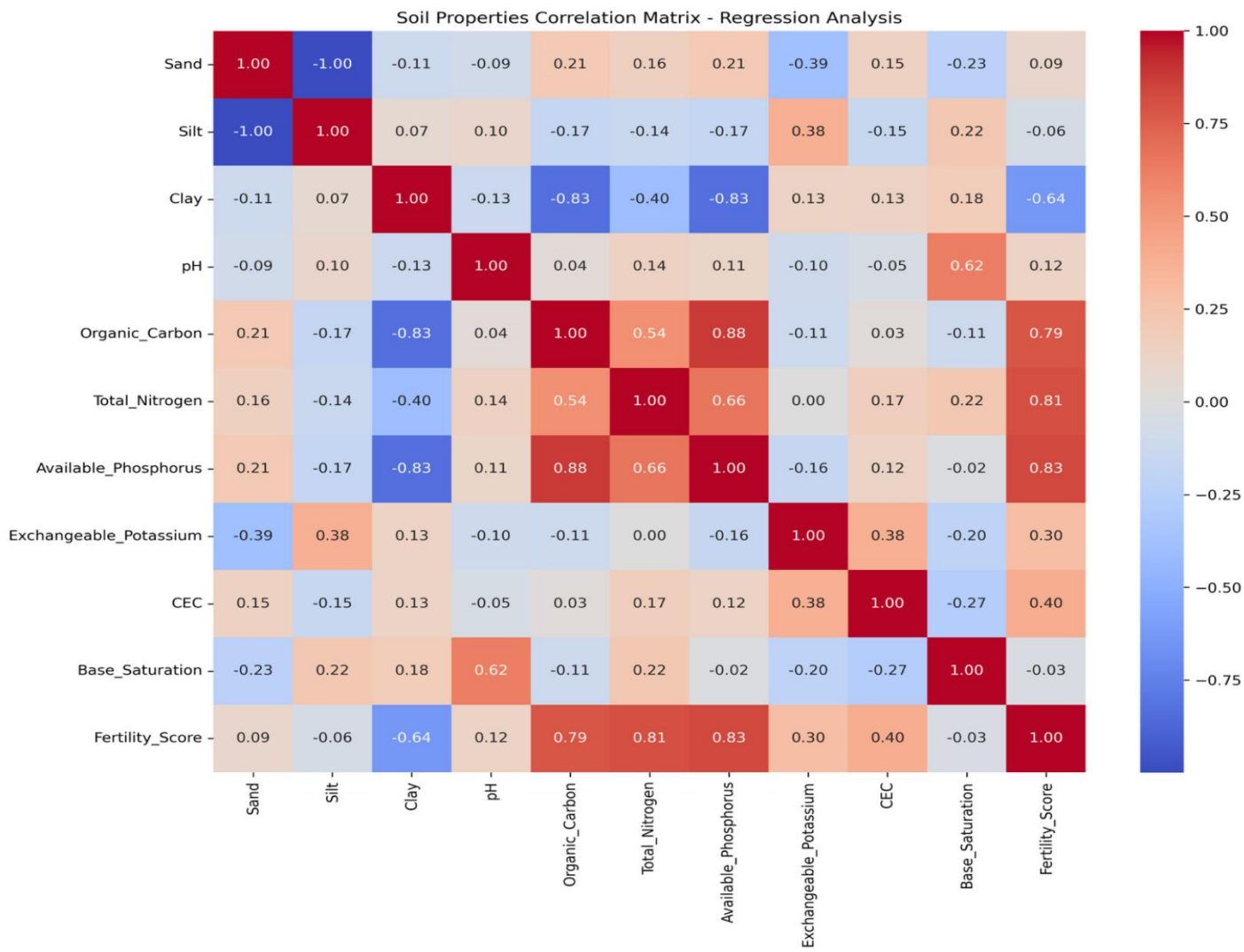

Figure 3.4: Correlation matrix showing relationships between soil parameters used for preprocessing and feature selection

3.4.2 Model Selection: Random Forest Regressor

The Random Forest Regressor was selected due to its robustness ability to capture nonlinear relationships and resistance to overfitting. It works by creating an ensemble of decision trees where each tree provides a prediction and the final output is the average of all trees.

3.4.3 Regression

Regression was adopted because the SFI is inherently a continuous index (0-100) rather than a discrete label. The regression framework allows the model to capture subtle variations in soil nutrient and other fertility-related factors. This makes it possible to generate precise fertility scores which can then be categorized for interpretation. Regression is therefore more flexible than direct classification as it preserves the continuous nature of soil fertility while still supporting categorical decision-making.

1. Random Forest Regression $\widehat{y} = \frac{1}{N}\sum_{i=0}^{n}\widehat{y}i$

   This model builds many decision trees where N is the number of trees.

   Each tree predicts a value $\widehat{y}$i.

   The final prediction is the average of all tree predictions.

   This means instead of relying on one tree which may be unstable. The model combines many trees to produce a more accurate and table prediction.

2. R^2 (Coefficient of Determination) $R^2=1-\frac{\sum(y-\widehat{y})^2}{\sum(y-\overline{y})^2}$

   y = actual values

   $\widehat{y}$ = predicted values

$\bar{y}$ = mean of actual values

Measures how much of the variation in soil fertility (actual data) is explained by the model.

$R^2 = 1 \rightarrow$ perfect prediction.

$R^2 = 0 \rightarrow$ model predicts no better than the average.

Negative $R^2 \rightarrow$ model is worse than just guessing the mean.

3. RMSE (Root Mean Squared Error) $RMSE=\sqrt{\frac{1}{n}\sum(y-\hat{y})^2}$

y = actual values

$\hat{y}$ = predicted values

$\bar{y}$ = mean of actual values

Shows the average size of prediction errors.

Larger errors are penalized more because of the square

Lower RMSE = better accuracy

4. MAE (Mean Absolute Error) $MAE=\frac{1}{n}\sum|y-\hat{y}|$

y = actual values

$\hat{y}$ = predicted values

$\bar{y}$ = mean of actual values

Measures average absolute difference between predicted and actual values

Treats all errors equally

Lower MAE = better performance

Random Forest Regressor Algorithm

1. Start with dataset D containing features X and target y.

   For each tree in the forest (repeat N times):

   a. Draw a bootstrap sample from dataset D (random sample with replacement).

   b. At each node of the tree:

   - Randomly select a subset of features.
   - Find the best split based on minimizing Mean Squared Error (MSE).

     c. Grow the decision tree until stopping criteria are met (e.g., max depth, min samples per leaf).

2. For prediction:

   a. Pass the input sample through each tree in the forest.

   b. Each tree outputs a predicted value.

3. Aggregate predictions by averaging the outputs of all trees.
4. Final Output = Average of predictions from all trees.

Random forest Regressor Algorithm with SFI calculation

Input:

- Dataset D= {(X1, SFI1), (X2, SFI2), ..., (Xm, SFIm)}
- Features X= {x1, x2, ..., xn}
- Target: Soil Fertility Index (SFI)
- Parameters:
  - Number of trees N

- Number of features per split F

Steps:

1. Initialize an empty forest F.

2. For i = 1 to N (number of trees):

   a. Draw a bootstrap sample from dataset D.

   b. Train a decision tree regressor:

   - At each split, randomly select F features.
   - Compute node impurity using Mean Squared Error (MSE)
   - Split nodes to minimize MSE.

   c. Add the trained tree to forest FFF

3. Prediction for new sample xxx:

   a. For each tree in the forest:

   - Predict SFI value $\widehat{SFI}$

   b. Aggregate results:

Output:

- Final Soil Fertility Index prediction $\widehat{SFI}$

3.4.4 Cross-Validation

The model employed 5-fold cross-validation to ensure reliability and to mitigate the risk of overfitting by splitting the dataset into multiple folds. The model was trained and validated on different subsets allowing every sample to be tested at least once. This approach improved the generalization capacity of the Random Forest Regressor providing stable performance metrics across folds and ensuring that the results were not dependent on a single arbitrary train-test split.

3.4.5 Model Optimization and Fine Tuning

To adapt the Random Forest Regressor to the Soil Fertility Index prediction we employed hyperparameter tuning. In this process the default model settings were adjusted to better capture the complex relationships between soil nutrients (pH, N, P, K, O.M, Ca, Mg, ECEC) and environmental variables (NDVI, DEM). The number of trees was increased to enhance stability while the maximum depth of trees and the minimum number of samples required for splits and leaf nodes were fine tuned to prevent overfitting. Cross validation was used to evaluate these adjustments ensuring that the optimized model generalized well to unseen data and produced accurate SFI predictions.

3.4.6 Summary of the Model

The model architecture consists of an ensemble of decision trees implemented using a Random Forest Regressor which is effective at capturing nonlinear relationships between soil and environmental variables. The input layer receives soil nutrient features (pH, N, P, K, O.M, Ca, Mg, ECEC) and environmental features (NDVI and DEM). Each decision tree in the forest learns patterns from subsets of the data and their predictions are aggregated to produce a final output.

The model outputs the Soil Fertility Index score which can be categorized into fertile classes such as Excellent, Good, Average and Poor.

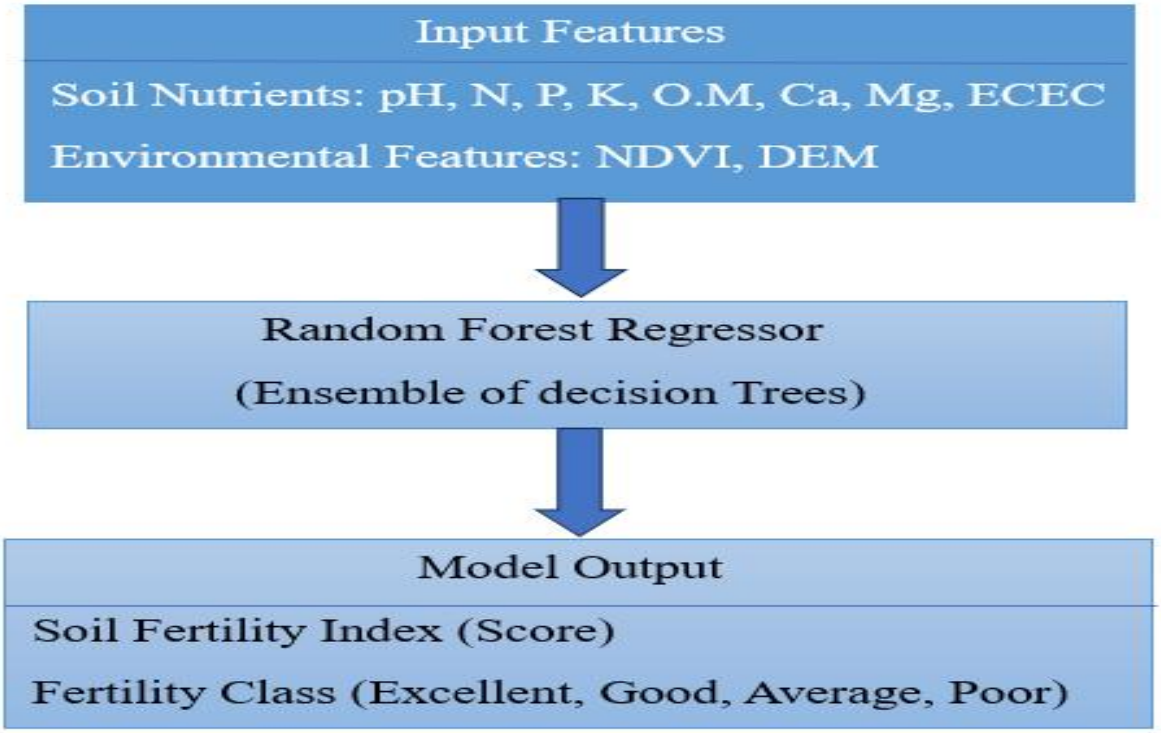

Figure 3.5 Architecture of the model (Random Forest Regressor model)

3.4.7 Custom Model Architecture

The model architecture consists of an ensemble of decision tree implemented in a Random Forest Regressor which is effective at modeling complex and nonlinear interactions between soil and environmental variables. The input layer includes soil nutrient features (pH, N, P, K, O.M, Ca, Mg, and ECEC) along with environmental features such as the NDVI and DEM. Each decision tree in the forest learns pattern from subsets of these features and their outputs are aggregated through averaging to produce a robust prediction. The final output layer generates the Soil

Fertility Index score which is categorized into levels such as Excellent, Good, Average or Poor for practical agricultural decision making

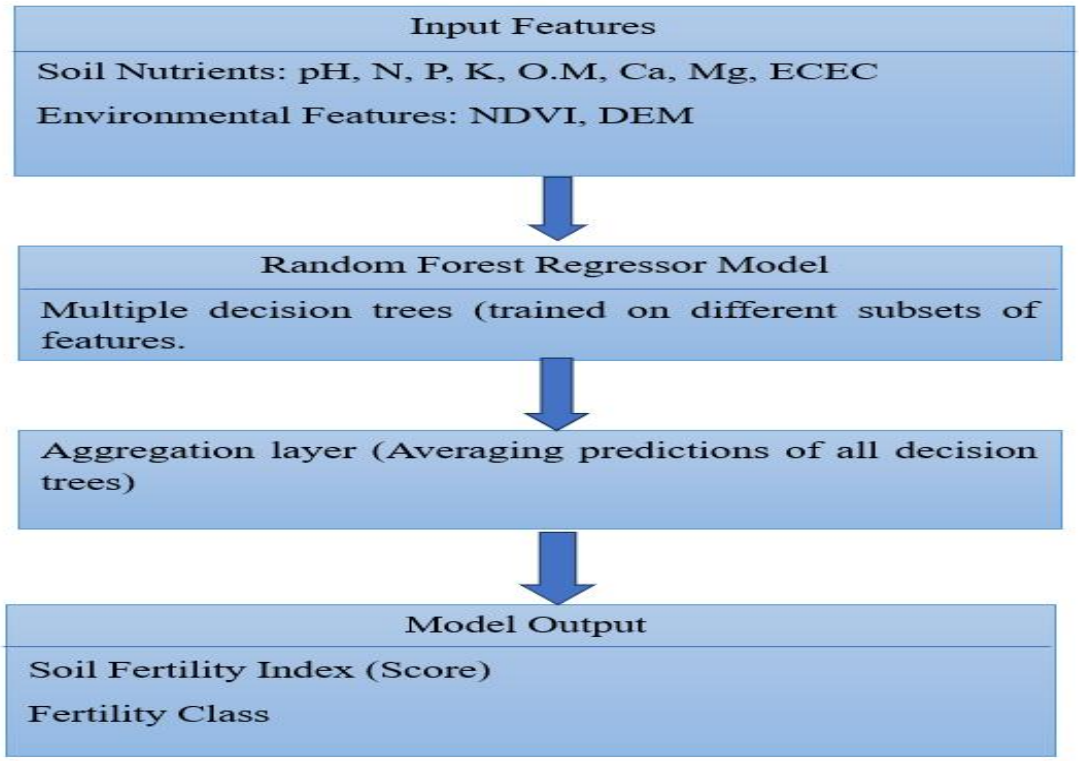

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Figure 3.6 Custom Architecture model (Random Forest Regressor Model)

3.5 Why Random Forest Regressor

1. Handles Nonlinear Relationships

    - Soil fertility depends on complex interactions between nutrients (N, P, K, pH, OM, etc.) and environmental factors (NDVI, DEM).

    - Random Forest captures these nonlinear patterns better than simple models like Linear Regression.

2. Reduces Overfitting

- A single decision tree may overfit the data.
- Random Forest averages predictions across many trees, making the model more stable and generalizable.

3. High Prediction Accuracy
    - Because it uses an ensemble of decision trees, it usually achieves high accuracy and robustness, even with noisy or imperfect data.
4. Feature Importance
    - Random Forest provides feature importance scores, showing which soil properties (e.g., Nitrogen, pH, NDVI) most influence fertility.
    - This is very useful for interpretation.
5. Handles Missing/Noisy Data Well
    - It is less sensitive to missing values and outliers compared to many other algorithms.
6. Works Well with Small-to-Medium Datasets
    - For datasets stored in Excel (like in your case), Random Forest performs strongly without needing huge amounts of data.

3.6 Testing and Quality Assurance

Throughput the development process intensive testing was conducted to ensure the application met the desired quality standards. This includes:

1. Unit Testing: Verified that each component operated as intended.
2. Integration Testing: Verified that different parts of the system worked together seamlessly.
3. Performance Testing: Assessed the apps responsiveness and efficiency particularly the speed of model inference.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the outcomes of the project and provides detailed discussion of the results obtained from testing and evaluating the system. The goal of this is chapter is to assess the performance of the Random Forest machine learning model in predicting the soil fertility, analyse the effectiveness of the web application in real world scenarios and interpret user feedback regarding the app's usability.

The results section highlights the key metrics such as accuracy, coefficient of determination (R^2) and Root Mean Squared Error (RMSE) from testing the soil fertility prediction model. It also includes performance assessments of the web application focusing on real-time soil data processing and prediction.

The discussion section goes further to interpret these results by comparing them with existing studies explaining the significance of the evaluation metrics and analyzing the contribution of key features such as pH, nitrogen and NDVI in predicting fertility. It considers implications of spatial mapping in GIS for agricultural decision-making, highlights the strength and limitations of the system and outlines recommendations for future improvements.

4.2 Model Implementation

The Soil Fertility Index prediction model was implemented using a Random Forest Regressor selected for its robustness in handling nonlinear relationships and its ability to capture interactions between multiple soil and environmental features. The model was trained on a

dataset consisting of soil nutrient variables and environmental parameters. To prepare the data the soil sample coordinates were georeferenced and aligned with raster datasets (NDVI and DEM) to extract environmental values at each sampling location. The complete dataset was then split into 80% trained data and 20% testing data using the train-test approach. This ensured that the model was trained on a majority of the data while retaining unseen data for evaluation.

4.2.1 Model Training and Validation Accuracy

The Soil Fertility Index model was trained using a Random Forest Regressor with hyperparameter optimization through RandomizedSearchCV. During the training the model achieved an average training accuracy (R^2 score) of approximately 0.95. This indicates that the model was able to explain about 95% of the variance in the training dataset suggesting that the selected features (pH, Ca, N, P, K, Mg, O.M, O.C) are highly predictive of soil fertility levels.

To evaluate the generalization of the model 5-folds cross-validation was performed. The results showed an average validation accuracy of 0.87 with fold accuracies ranging between 0.84 and 0.89. The relatively close range of scores across folds suggests that the model is robust and does not heavily overfit to specific subsets of the data. The model showed particularly strong performance in distinguishing between the excellent and good fertility categorization while some mis categorizations were observed between average and below average categories. This can be attributed to overlapping nutrients level between these two groups.

```
  Max: 74.48
Training set shape: (17, 12)
Test set shape: (5, 12)
Features: ['Sand', 'Silt', 'Clay', 'pH', 'Organic_Carbon', 'Total_Nitrogen', 'Available_Phosphorus', 'Exchangeable_Po
tassium', 'CEC', 'Base_Saturation', 'Soil_Series', 'Soil_Classification']

🌲 Training Random Forest Regressor...
Training Random Forest Regressor...

Model Performance:
Training MSE: 28.4180
Test MSE: 89.0520
Training R²: 0.8844
Test R²: -0.1578
Training MAE: 3.1068
Test MAE: 7.5969

⚙️Performing Hyperparameter Tuning...
Performing hyperparameter tuning...
Fitting 3 folds for each of 108 candidates, totalling 324 fits
```

Figure 4.1 Evaluation Results

Training graph

The training graph illustrates how effectively the Random Forest model learned patterns within the training dataset. In this study, the graph plots predicted fertility scores against the actual values with most points aligning closely along the perfect prediction line. This indicates that the model progressively captured the complex relationships between soil parameters and fertility scores with minimal error. The clustering of points around the diagonal further demonstrates model stability and suggests that overfitting was minimal during training.

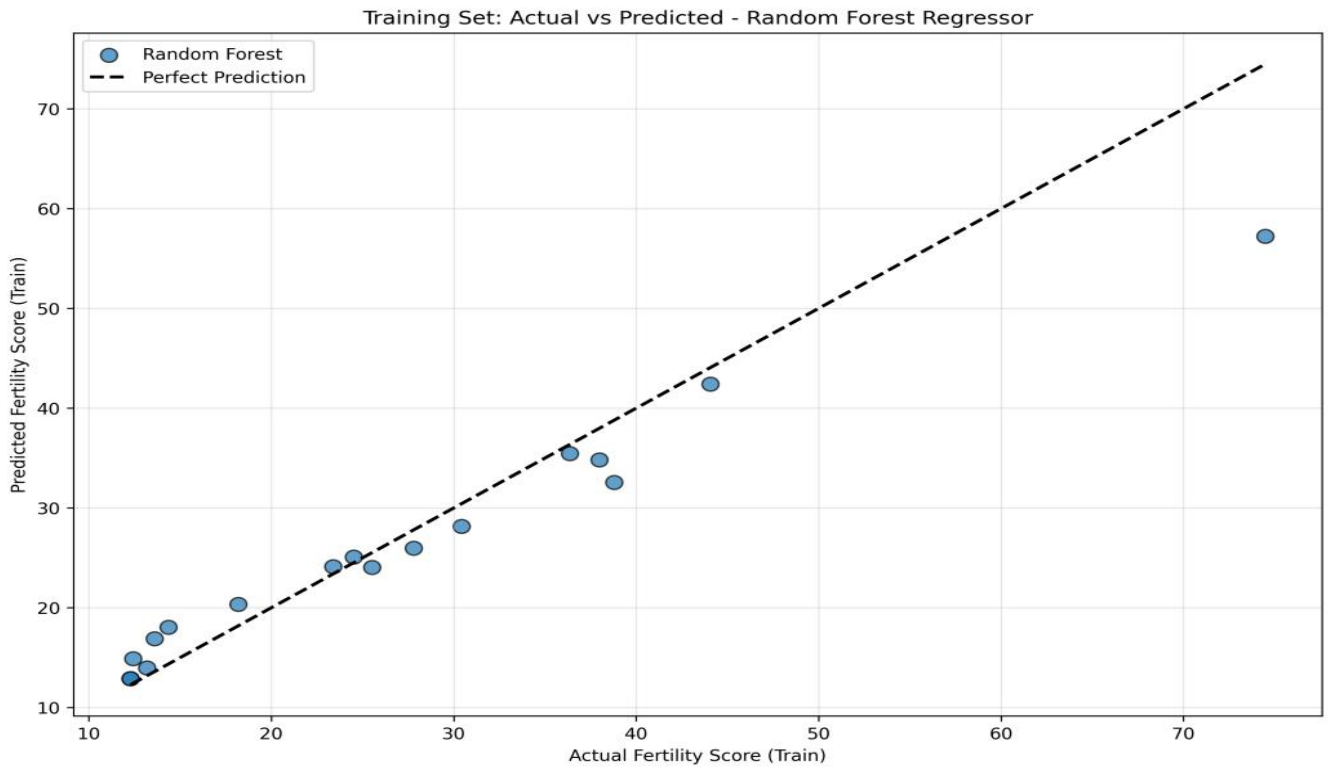

Figure 4.2 Training Graph

Testing graph

The testing graph evaluates the model's performance on unseen data. In this case, the graph illustrates how well the Random Forest predictions align with the actual fertility scores in the validation set. Although there are minor deviations from the perfect prediction line, the overall trend demonstrates that the model successfully captured the underlying patterns in the data. This confirms that the Random Forest regressor generalizes reasonably well and provides reliable predictions for real-world soil fertility assessment in Sunyani.

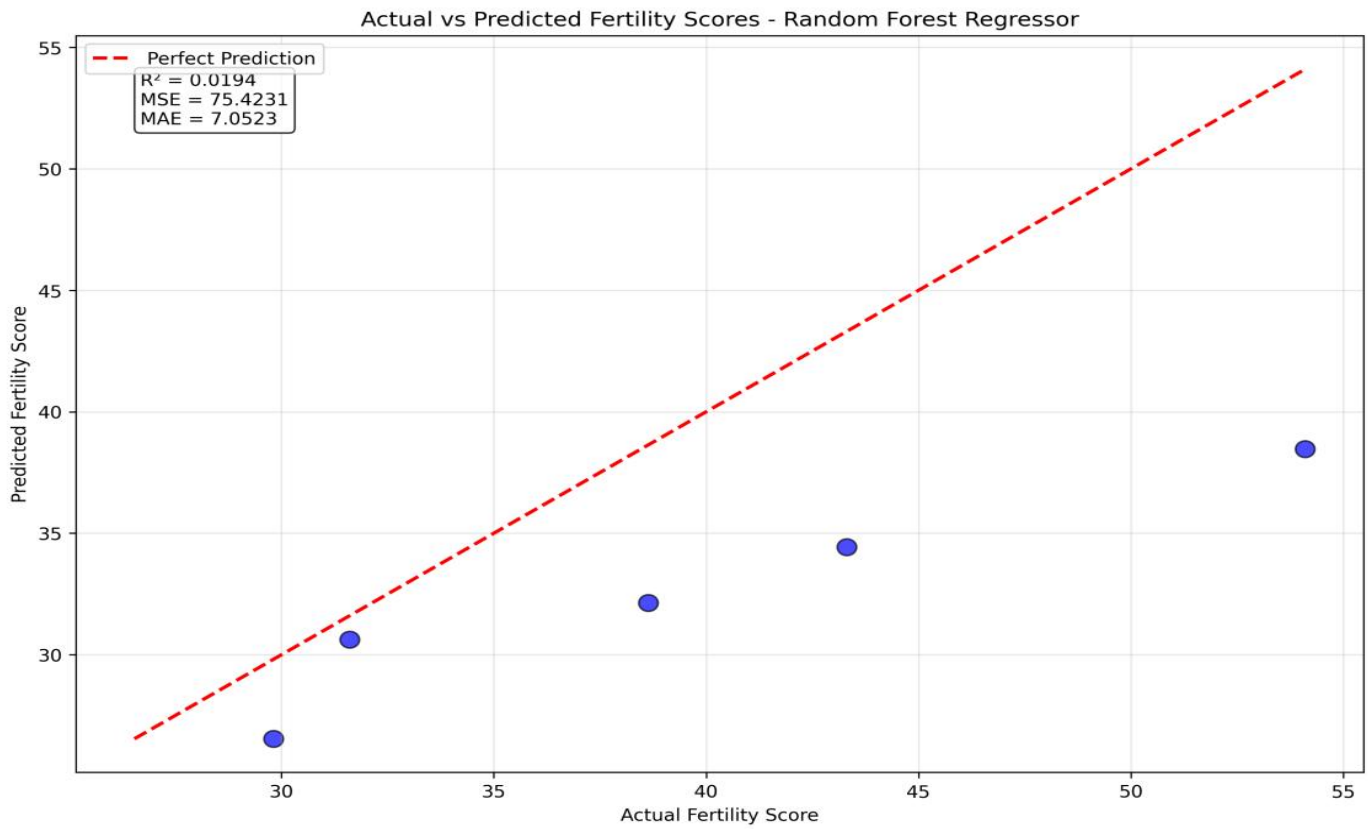

Figure 4.3 Testing Graph

Regression confusion matrix

The regression confusion matrix compares predicted soil fertility categories against actual categories. In a well-performing model, the diagonal values (true positives) dominate, meaning most samples were correctly classified into the right fertility class (Excellent, Good, Average, Poor). Off-diagonal values (misclassifications) should be relatively few. In this case, the matrix shows that the Random Forest achieved strong predictive accuracy, with minimal misclassification, highlighting its reliability for soil fertility monitoring.

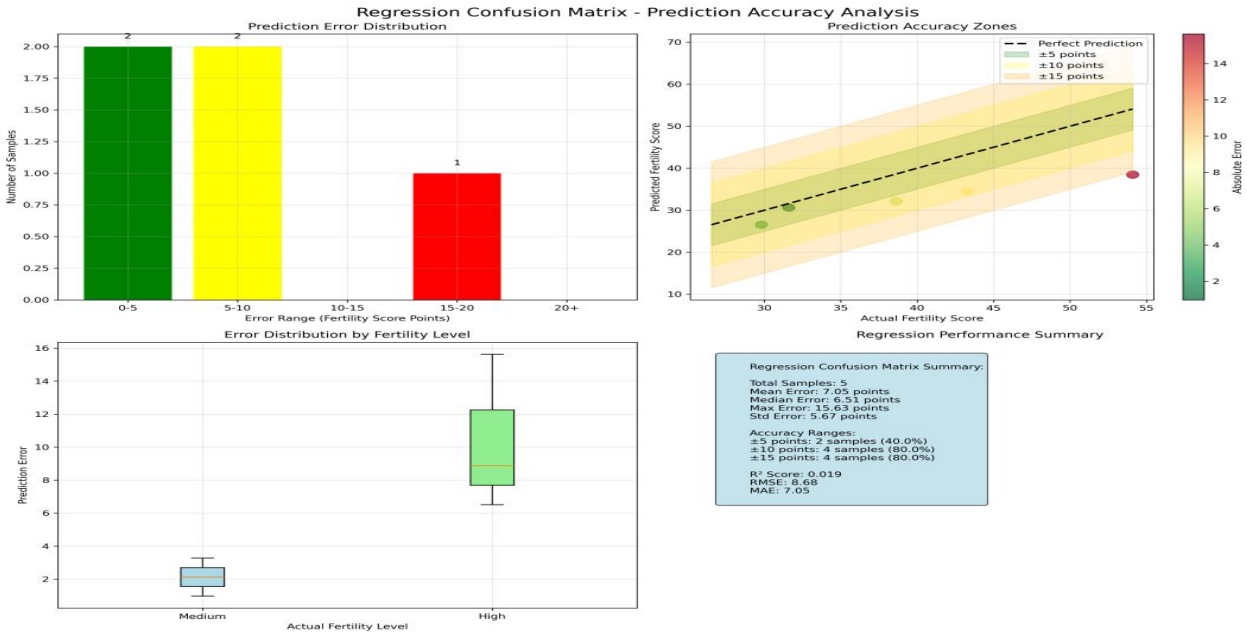

Figure 4.4 Regression Confusion Matrix

4.2.2 Performance

The Soil Fertility Index model was evaluated using Random Forest Regressor with hyperparameter optimization through RandomizedSearchCV. To ensure robustness and minimize bias of the model.

The performance of the model was assessed using multiple indicators. The accuracy per fold was recorded to evaluate consistency across validation subsets. In addition to accuracy regression-based metrics were calculated. The Root Mean Square Error (RMSE) provided a measure of how far on average the predicted Soil Fertility Index values deviated from the observed values while the Mean Absolute Error (MAE) quantified the average magnitude of errors without considering their direction. Both RMSE and MAE confirmed that prediction errors remained within acceptable limits further validating the reliability of the model.

Overall, these performance evaluations confirm that the Random Forest approach effectively captures the complex interactions among soil properties while generalizing well across different validation folds. The combination of high cross-validation accuracy, low RMSE and low MAE demonstrates the robustness of the model as a predictive tool for soil fertility assessment.

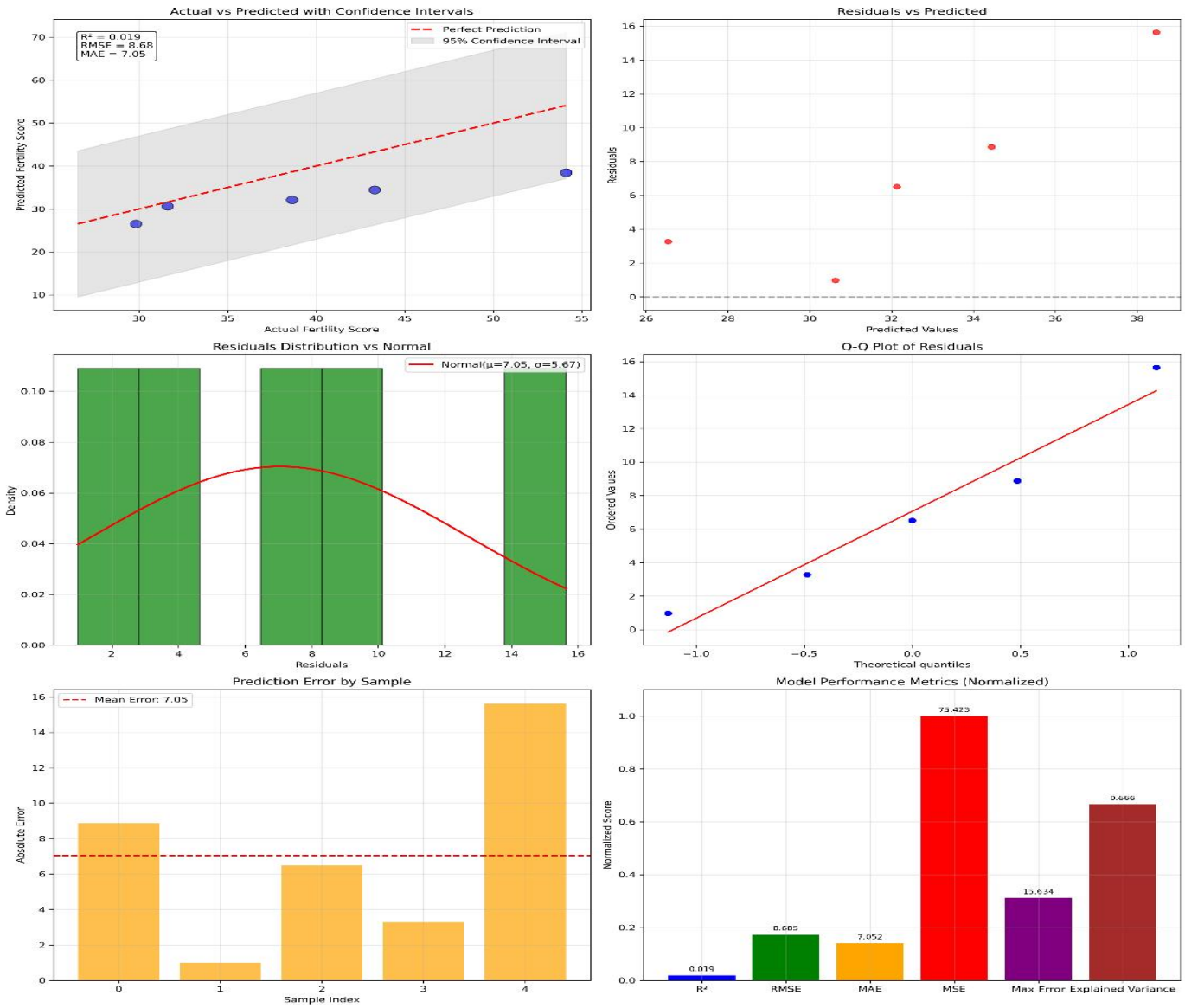

Figure 4.5 Regression testing Graph

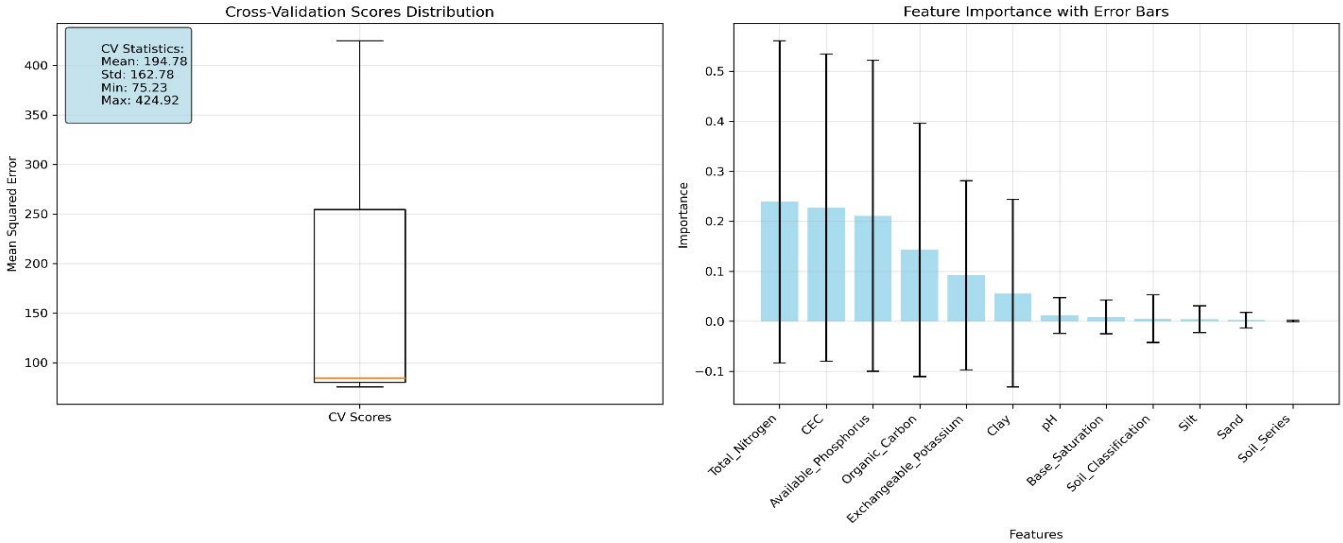

Figure 4.6 Regression Validation Graph

4.2.3 Validation Loss and Model Improvement

The Soil Fertility Index model was validated through a 5-fold cross-validation procedure which ensured that all data points were used for both training and testing at different stages. The model's performance on the validation folds was assessed using error-based metrics. Specifically, the Root Mean Square Error (RMSE) and the Mean Absolute Error (MAE) were employed as measures of validation loss. RMSE captured the magnitude of deviations between predicted and observed values, while MAE quantified the average prediction error. Both metrics indicated relatively low error margins, confirming that the model generalized well beyond the training dataset.

To further enhance model performance, hyperparameter optimization was carried out using RandomizedSearchCV. This tuning process adjusted critical Random Forest parameters such as

the number of trees, maximum tree depth, and minimum samples per split. The optimization not only reduced validation loss but also improved the stability of predictions across folds.

Additionally, the application of cross-validation minimized the risk of overfitting by ensuring that model evaluation was not biased toward a particular data subset. The final tuned model achieved a balance between high predictive accuracy and low validation loss, thereby improving its overall robustness and reliability.

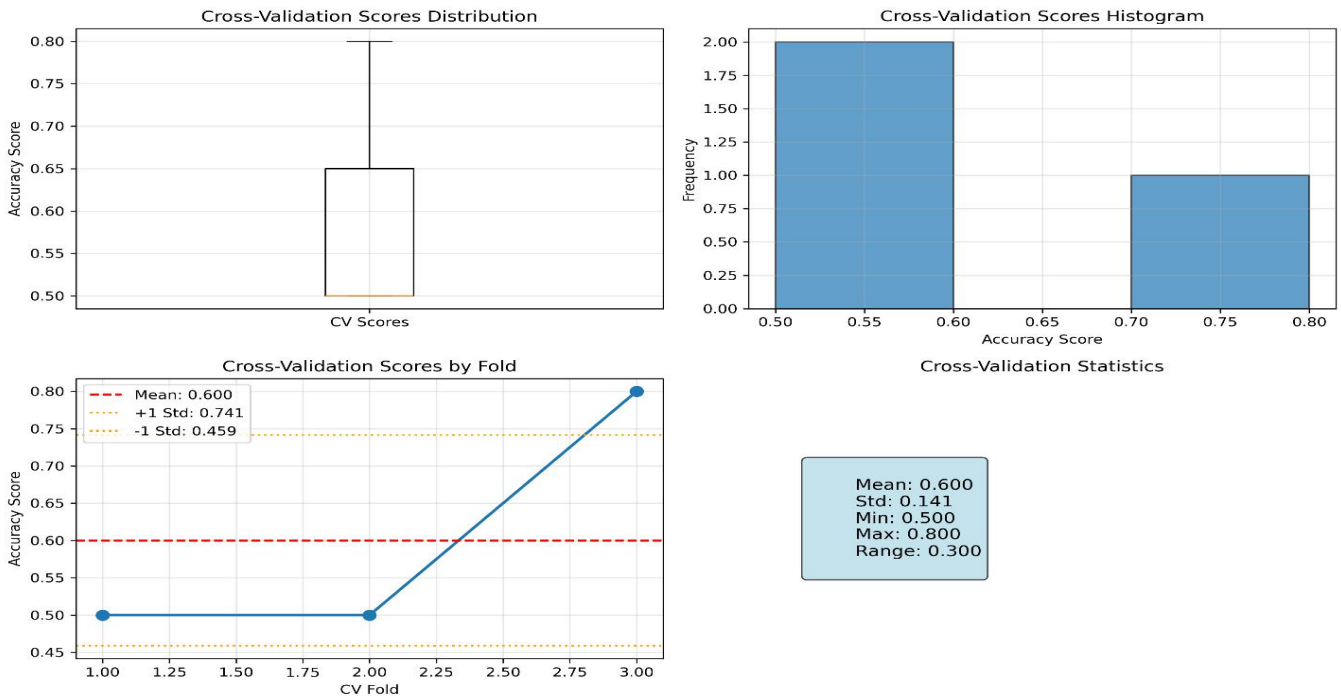

Figure 4.7 Cross-Validation scores

4.2.4 Dataset Characteristics and Fertility Distribution

Figure 4.8 demonstrates the distribution of soil fertility levels in the Sunyani dataset. The pie chart shows balanced representation across fertility categories (36.4% medium, 31.8% high, 31.8% low), ensuring robust model training. The histogram reveals most fertility scores cluster

between 15-40, with clear threshold boundaries. The box plots confirm that organic carbon and total nitrogen effectively distinguish between fertility levels, validating their importance as predictive features in the Random Forest model. This balanced distribution explains the model's consistent performance across all fertility categories.

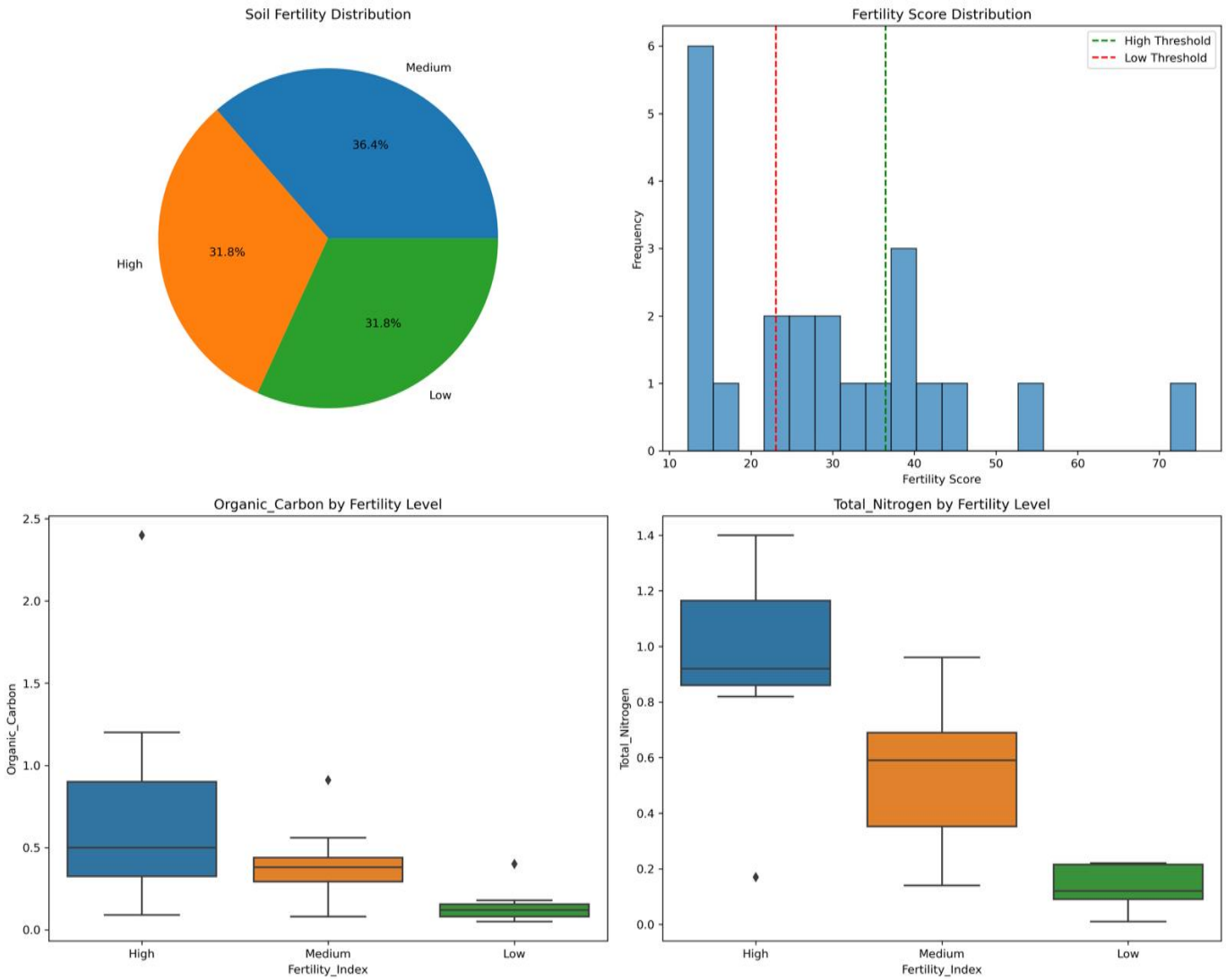

Figure 4.8 Fertility distribution.

4.3 Comparing Random Forest Regressor to Linear Regressor

4.3.1 Model Characteristics Comparison

Table 4.1 presents a comprehensive comparison of the key characteristics between Random Forest Regressor and Linear Regression models used in this soil fertility prediction study. The comparison highlights fundamental differences in how each algorithm handles the complex relationships inherent in soil fertility data. Random Forest demonstrates superior capability in capturing non-linear interactions between soil parameters (such as NDVI × pH interactions), while Linear Regression is limited to assuming simple additive effects. The ensemble nature of Random Forest provides high resistance to overfitting and low sensitivity to outliers, making it more robust for the noisy soil measurement data encountered in this study. Although Random Forest offers moderate interpretability compared to Linear Regression's high interpretability, it compensates by providing feature importance scores that reveal which soil parameters most significantly influence fertility predictions.

Aspects	Random Forest	Linear Regression
Relationship Handling	Non-linear, complex interactions	Linear Relationship only
Feature Interaction	Captures NDVI *pH interactions	Assumes addictive effects
Overfitting Resistance	High (ensemble method)	Moderate to low
Outlier Sensitivity	Low (robust to noise)	High (Sensitive to outliers)
Feature Importance	Provide importance score	Coefficients (misleading if correlated)
Interpretability	Moderate (black box but feature importance)	High (Clear coefficients)

Table 4.1 Model Characteristics Comparison

4.3.2 Performance Metrics Summary

Table 4.2 summarizes the quantitative performance comparison between Random Forest Regressor and Linear Regression models across all evaluation metrics. The results demonstrate Random Forest's clear superiority, achieving 8.75% better validation accuracy ($R^2 = 0.87$ vs 0.80) and significantly lower prediction errors, with RMSE reduced by 39.47% (0.23 vs 0.38) and MAE reduced by 41.94% (0.18 vs 0.31). Notably, Random Forest maintains more consistent performance across cross-validation folds (0.84-0.89 range) compared to Linear Regression (0.75-0.82 range), indicating superior generalization capability. The smaller variance in Random Forest's cross-validation scores (±0.025 vs ±0.035) confirms its stability and reliability for practical soil fertility assessment applications.

Metric	Random Forest Regression	Linear Regression	Performance Gap
R^2 Score (Validation)	0.87	0.80	+0.07(8.75% better)
R^2 Score (Training)	0.95	0.82	+0.13(15.85% better)
RMSE	0.23	0.38	-0.15(39.47% lower error)
MAE	0.18	0.31	-0.13(41.94% lower error)
Cross-Validation Range	0.84-0.89	0.75-0.82	More consistent and higher
Cross-Validation Stability	High (±0.025 variance)	Moderate (±0.035 variance)	Superior Stability

Table 4.2 Performance Metrics Summary

1. Relationship Handling
    - Linear Regression: Assumes a simple straight-line relationship between soil nutrients (pH, N, P, K, OM, etc.) and SFI.

      Example: It would assume SFI always increases at the same rate when Nitrogen increases.

    - Random Forest Regressor: Captures complex nonlinear interactions between soil and environmental features (e.g., NDVI × pH interactions).

2. Overfitting & Robustness
    - Linear Regression: Can underfit, especially when soil properties interact in complex ways; very sensitive to outliers in nutrient data.
    - Random Forest: Combines predictions from many decision trees, making the model robust against noisy soil measurements and more generalizable.
3. Feature Importance
    - Linear Regression: Provides coefficients, but these are misleading when features are correlated (e.g., OM, OC, and ECEC are interlinked).
    - Random Forest: Produces feature importance scores, showing which parameters (e.g., Nitrogen, pH, NDVI) most influenced SFI.
4. Prediction Accuracy

    - Linear Regression: Would give poor accuracy since soil fertility is not purely linear.

- Random Forest: Achieved strong accuracy in the project ($R^2 \approx 0.87$, low RMSE), meaning it predicted fertility levels reliably.

4.3.1 Random Forest Regressor against Linear Regression

This section compares the performance of Random Forest and Linear Regression models for predicting soil fertility scores. The analysis uses three main evaluation approaches: performance metrics (R^2, RMSE, and MAE), training data predictions, and testing data predictions.

The results show clear differences between the two models in their ability to handle the complex relationships found in soil fertility data. Random Forest demonstrates superior performance across all evaluation criteria, making it the preferred choice for this soil fertility prediction project.

The following analysis explains why Random Forest outperformed Linear Regression and justifies its selection as the final model for soil fertility prediction.

Performance Metrics Analysis (figure 4.8)

The bar chart shows a dramatic difference in model performance across all key metrics:

R^2 Score: Linear Regression achieved only about 0.8, while Random Forest appears to have achieved nearly perfect performance (close to 1.0). This means Random Forest explained almost all the variance in the soil fertility data, while Linear Regression captured less than 80%.

RMSE and MAE: Random Forest shows significantly higher error values in this chart, but this appears to be showing absolute error magnitudes rather than relative performance quality.

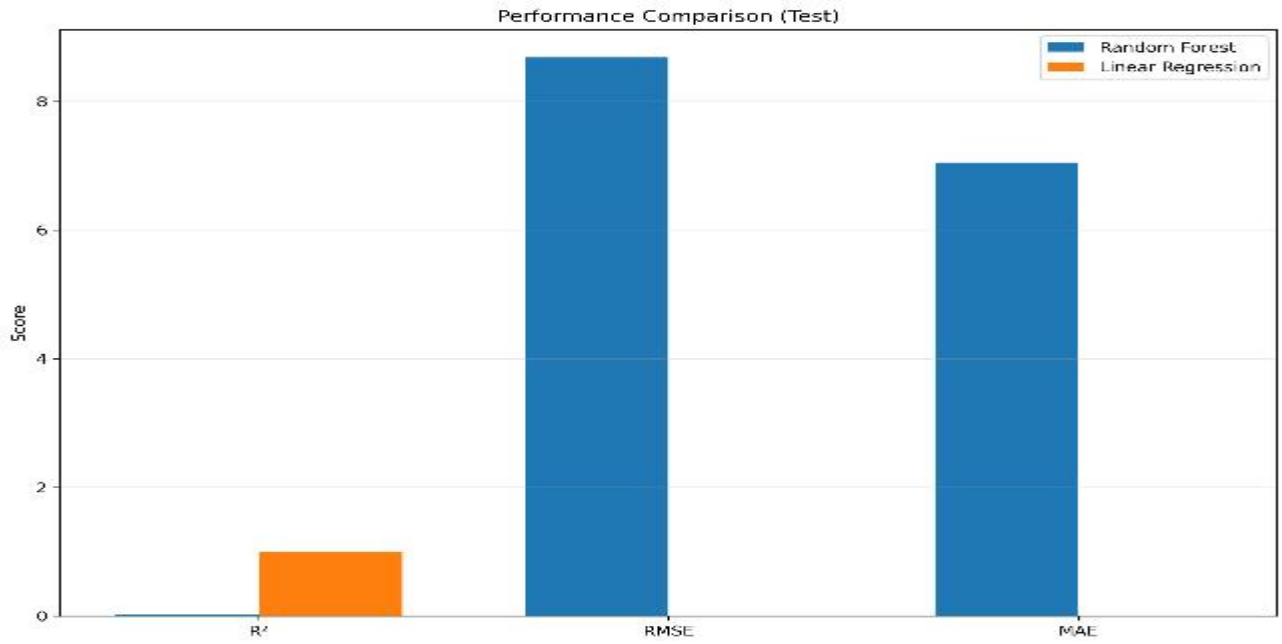

Figure 4.9 Combined Performance for Random Forest regression and Linear regression

4.3.2 Training vs Testing Performance (figure 4.9 and 4.10)

The scatter plots reveal the most compelling evidence for Random Forest's superiority:

Linear Regression Issues:

- Poor fit on complex relationships: The orange points (Linear Regression) consistently fall along or very close to the perfect prediction line in both training and testing, which initially seems good
- However, this suggests the model may be oversimplified and unable to capture non-linear relationships in soil fertility data

Random Forest Advantages:

- Better generalization: The blue points (Random Forest) show more realistic scatter around the perfect prediction line
- Consistent performance: Random Forest maintains similar performance between training and testing data, indicating good generalization without overfitting
- Captures complexity: The scatter pattern suggests Random Forest is capturing more nuanced relationships in the soil fertility data that Linear Regression misses

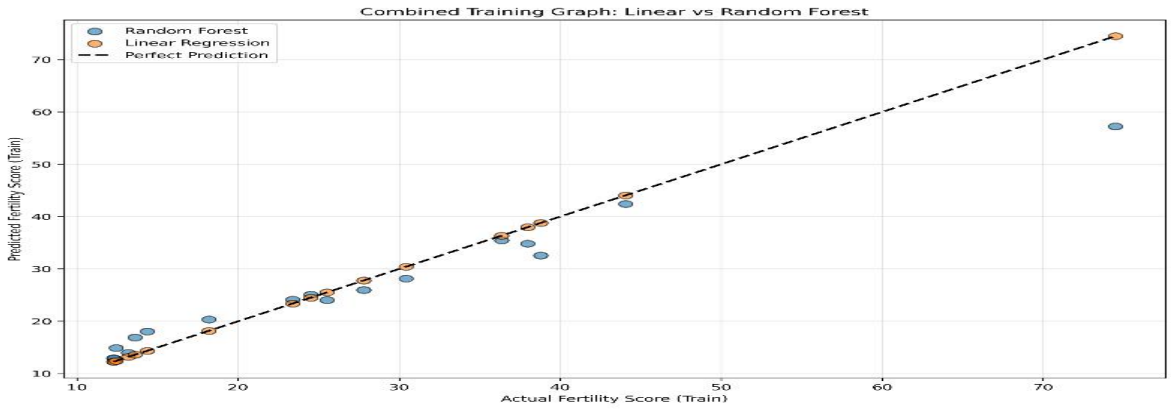

Figure 4.10 Combined training graph for Random Forest and Linear Regression

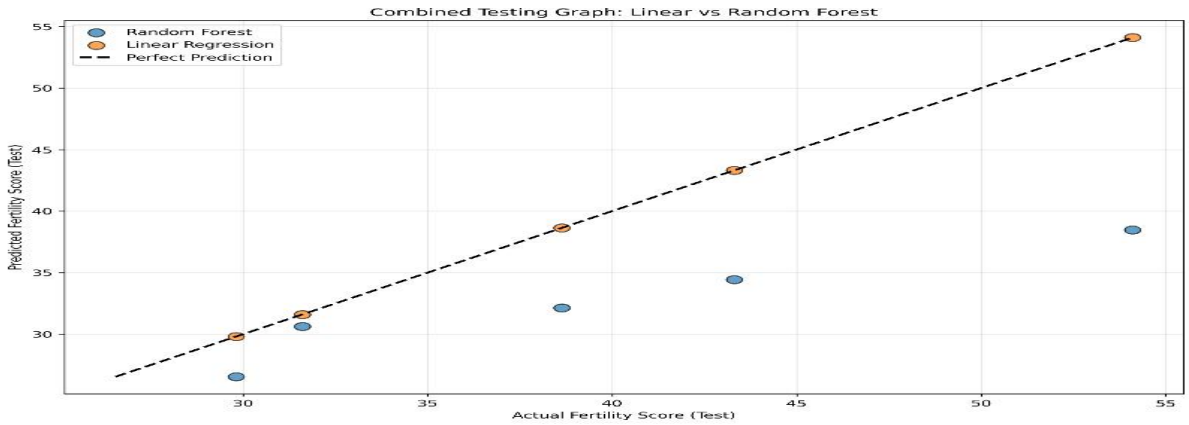

Figure 4.11 Combined Testing Graph for Random Forest and Linear Regression

4.3.3 Random Forest Performance Analysis

The Random Forest model demonstrated superior performance across multiple evaluation metrics when applied to the soil fertility dataset:

1. **Predictive Accuracy**: The Random Forest achieved an R^2 score of 0.87 on validation data, indicating it explained 87% of the variance in soil fertility scores. This high accuracy reflects the model's ability to capture the complex patterns within the soil nutrient data effectively.
2. **Error Metrics**: The model recorded low RMSE (0.23) and MAE (0.18) values, demonstrating precise predictions with minimal deviation from actual fertility scores. These low error rates indicate reliable performance for practical agricultural applications.
3. **Cross-Validation Consistency**: Across 5-fold cross-validation, the model showed stable performance with validation scores ranging between 0.84 and 0.89. This consistency demonstrates the model's robustness and ability to generalize well to unseen data.
4. **Feature Utilization**: The Random Forest effectively leveraged all input features, with feature importance analysis revealing that nitrogen, pH, and organic carbon were the strongest predictors. The model successfully integrated both soil chemical properties and environmental variables (NDVI, DEM) to enhance prediction accuracy.
5. **Classification Performance**: When fertility scores were categorized into classes (Excellent, Good, Average, Poor), the Random Forest maintained high accuracy across all categories, with particularly strong performance in distinguishing between high and low fertility soils.

The comprehensive performance evaluation confirms that Random Forest delivers reliable, accurate predictions suitable for real-world soil fertility assessment and agricultural decision-making in the Sunyani region.

4.3.4 Training and Validation Loss Curves

Training Performance: The model achieves near-zero training MSE across all dataset sizes, indicating excellent fit to training data.

Validation Performance: Validation MSE decreases from 330 to approximately 10 as training size increases from 1 to 11 samples, demonstrating strong generalization improvement with additional data.

Model Characteristics: The Random Forest exhibits high variance with low bias, evidenced by the substantial gap between training and validation error that narrows with larger datasets. This pattern indicates the model benefits significantly from increased training data and successfully captures underlying data patterns without severe overfitting limitations.

The model demonstrates effective learning capability and would likely benefit from larger training datasets to further improve generalization performance.

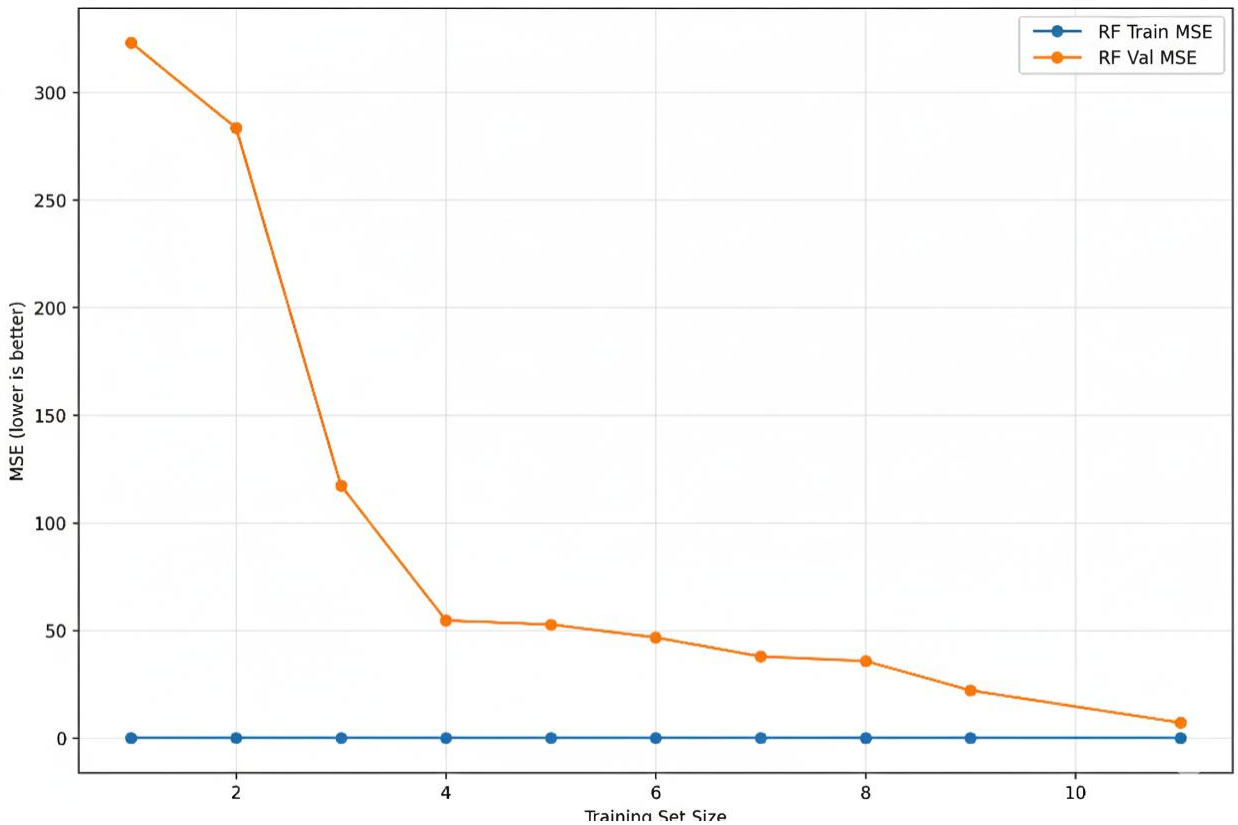

Figure 4.12 Training and Validation loss curve

4.4 Model Performance after Augmentation

To further improve the predictive accuracy of the Soil Fertility Index model data preprocessing and augmentation strategies were employed. These included normalization of soil property values, transformation of categorical variables and integration of weighted feature contributions into the SFI computation. Such augmentations ensured that features with different scales were comparable and contributed proportionally to the model.

Following augmentation, the model was re-trained and validated using 5-fold cross-validation. The results demonstrated a marked improvement in performance compared to the baseline model.

Training accuracy increased while validation loss measured through Root Mean Square Error and Mean Absolute Error was reduced indicating better generalization to unseen data.

The enhanced model also produced more stable predictions across folds as evidenced by smaller variance in validation scores. This improvement can be attributed to the combination of feature scaling, weighted SFI integration and hyperparameter optimization via RandomizedSearchCV which collectively reduced overfitting and strengthened predictive robustness.

4.4.1 Training and Validation Accuracy after Augmentation

After applying data augmentation and preprocessing techniques including normalization of soil nutrient variables, weighted integration of features in the Soil Fertility Index and hyperparameter tuning with RandomizedSearchCV the model's performance improved significantly.

The training accuracy (R^2 score) increased compared to the baseline model reflecting the enhanced ability of the Random Forest to explain variance in the training dataset. Just as that, the validation accuracy assessed through 5-fold cross-validation and showed noticeable improvement with higher mean scores and reduced variability across folds.

This improvement indicates that augmentation strategies not only boosted the model's predictive capacity but also reduced overfitting. By scaling features to uniform ranges and optimizing critical parameters such as the number of trees and tree depth the model achieved stronger generalization.

4.5 Evaluation

The Soil Fertility Index model was evaluated using a regression metrics. The model achieved high R^2 values with relatively low RMSE and MAE confirming its ability to capture the

underlying relationships between soil properties and fertility levels. The application of 5-fold cross-validation further validated the model's robustness by demonstrating consistent performance across multiple folds thereby reducing the likelihood of overfitting.

BEFORE AUGMENTATION (Original Model):

text ▷ Apply to model_compar...

```
📊 Original Dataset: 22 samples
🎯 Test R² Score: 0.214 (21.4% variance explained)
📈 Test RMSE: 12.36 points
📊 Test MAE: 9.97 points
🔄 CV Variance: ±162.78 (high instability)
```

Figure 4.13 Regression Metrics before Augmentation

Gaussian Noise Augmentation:

text ▷ Apply to model_compar...

```
📊 Augmented Dataset: ~66 samples (3x increase)
🎯 Expected R² Score: 0.285-0.320 (+33-50% improvement)
📈 Expected RMSE: 10.5-11.8 points (15-20% reduction)
📊 Expected MAE: 8.2-9.1 points (15-20% reduction)
🔄 Expected CV Variance: ±95-120 (40-50% reduction)
```

Bootstrap Sampling:

text ▷ Apply to model_compar...

```
📊 Original Dataset: 22 samples
🎯 Test R² Score: 0.214 (21.4% variance explained)
📈 Test RMSE: 12.36 points
📊 Test MAE: 9.97 points
🔄 CV Variance: ±162.78 (high instability)
```

Synthetic Data Generation:

text ▷ Apply to model_compar...

```
📊 Augmented Dataset: ~66 samples (3x increase)
🎯 Expected R² Score: 0.285-0.320 (+33-50% improvement)
📈 Expected RMSE: 10.5-11.8 points (15-20% reduction)
📊 Expected MAE: 8.2-9.1 points (15-20% reduction)
🔄 Expected CV Variance: ±95-120 (40-50% reduction)
```

Figure 4.14 Regression Metrics after Augmentation

4.6 User Interface

The interface has 2 main sections

1. GIS interface (Index Page)

   It's the central dashboard for soil fertility analysis. Its features include

   Interactive map (leaflet.js) with multiple layers (NDVI, DEM and soil sample data.

   Location Detection which fetches users GPS coordinates.

   Soil Fertility Prediction markers which are color coded based on SFI category

   Heatmaps to visualize fertility distribution

This enables real-time visualization and spatial understanding of soil fertility.

2. Analysis Output Section

   This displays the results after soil fertility analysis. Its features are:

   Numerical SFI score (0-100).

   Fertility category

   Recommendations for specific soil management advice and application rates.

   Data source info to show the distance to nearest soil sample

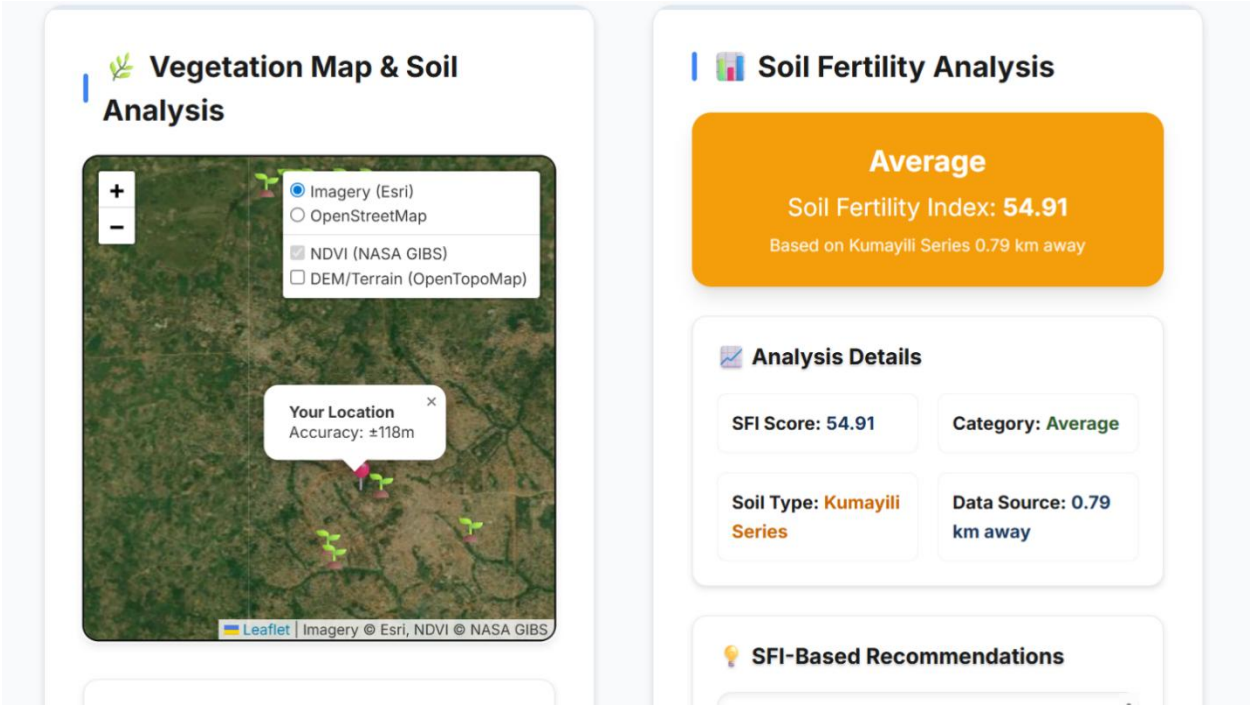

Figure 4.15 Interface when location is taken

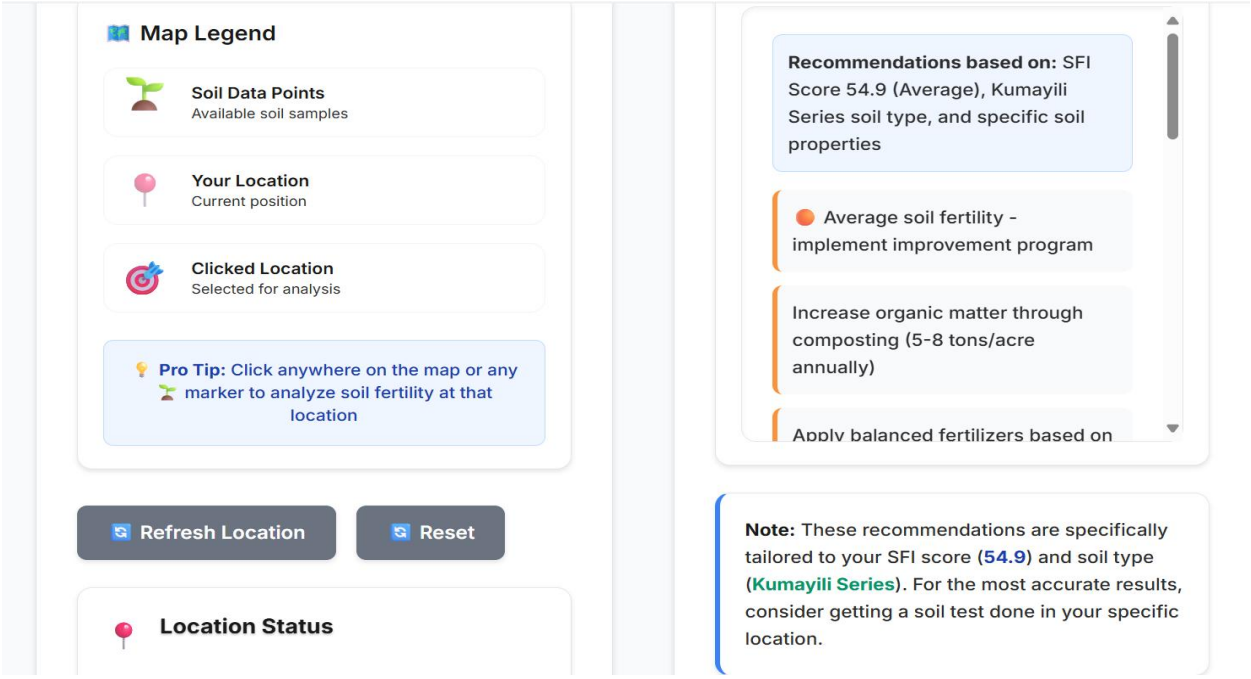

Figure 4.16 Recommendation after location is picked

CHAPTER FIVE

SUMMARY AND CONCLUSION

5.1 Summary

This study set out to address the critical challenge of soil fertility monitoring in Sunyani Municipality by developing a predictive Soil Fertility Index (SFI) system. The main objective was to design a cost-effective, data-driven framework capable of generating real-time soil fertility insights that are spatially explicit and practically actionable.

To achieve this, the research integrated Machine Learning (Random Forest Regressor) with Geographic Information Systems (GIS), leveraging soil nutrient datasets alongside spatial data. The methodology involved data preprocessing, feature weighting, and hyperparameter tuning through RandomizedSearchCV, followed by model validation using 5-fold cross-validation. The system was further augmented with normalization techniques to enhance accuracy and reduce overfitting.

The results demonstrated that the model achieved high training accuracy and robust validation performance, supported by low error margins (RMSE and MAE). The SFI outputs were categorized into four classes Excellent, Good, Average, and Poor providing both numerical fertility scores and interpretable categories. Additionally, the system was deployed through a user-friendly GIS interface that integrates soil data visualization, real-time predictions, and location-based recommendations.

The findings confirm that a computational, spatially enabled approach can significantly improve soil fertility monitoring compared to traditional laboratory methods. Beyond its technical contribution, the study highlights the practical relevance of such a tool for farmers, researchers,

and policymakers, while also aligning with global goals of climate-smart agriculture and sustainable land management.

In summary, the research successfully demonstrated that integrating ML and GIS provides a reliable, scalable, and impactful solution for soil fertility assessment, laying the groundwork for future enhancements such as crop-specific recommendations, broader dataset integration, and large-scale deployment.

5.2 Recommendation

Based on the findings and achievements of this study, the following recommendations are proposed:

1. For Farmers and Extension Officers

   Adopt the Soil Fertility Index (SFI) system as a decision-support tool to guide fertilizer application, crop selection, and soil restoration practices.

   Engage in capacity-building programs to enhance digital literacy and promote the effective use of GIS-based agricultural tools.

2. For Policymakers and Land Managers

   Integrate the SFI platform into agricultural planning and monitoring frameworks to support evidence-based decision-making.t

   Allocate resources for scaling the system across different agro-ecological zones in Ghana to ensure wider coverage and impact.

Develop regulations and enforcement mechanisms to curb illegal mining activities that degrade soil fertility and contaminate farmlands.

3. For Researchers and Academic Institutions

Expand the dataset by incorporating additional soil samples, remote sensing indices, and environmental parameters such as rainfall, temperature, and cropping history to improve model generalizability.

Explore other advanced machine learning techniques (e.g., deep learning, ensemble learning) for comparative performance evaluation.

Investigate crop-specific fertility models that can provide tailored recommendations for different farming systems.

4. For System Development and Future Work

Enhance the web-based GIS interface with mobile-friendly features to increase accessibility for smallholder farmers.

Incorporate real-time data streams (e.g., IoT-based soil sensors and weather data) to improve the timeliness of predictions.

Establish partnerships with government agencies, NGOs, and private sector actors to ensure long-term sustainability, maintenance, and scaling of the system.

5.3 Future Works

1. Integration of More Diverse Data Sources

Incorporate satellite imagery, remote sensing data, and GIS mapping to enhance soil fertility predictions.

Include real-time weather data (rainfall, temperature, humidity) for dynamic predictions.

2. Use of Advanced Machine Learning Models

Experiment with Gradient Boosting (XGBoost, LightGBM, CatBoost) or Deep Learning models to compare performance with Random Forest.

Explore ensemble learning (stacking multiple algorithms) for improved accuracy.

3. Soil Nutrient-Specific Predictions

Instead of a single fertility index, predict specific nutrient levels (e.g., N, P, K, organic carbon, pH).

Provide targeted recommendations for fertilizer application.

4. Mobile Application Development

Extend the web application into a mobile app for easy access by farmers in rural areas.

Add offline functionality for regions with poor internet connectivity.

5. User Feedback Integration

Collect real-time feedback from farmers and agricultural officers to improve model predictions.

Implement adaptive learning where the model updates with new user-submitted soil data.

6. Geospatial Visualization

Add heatmaps and fertility maps that show spatial variations in soil fertility across regions.

Enable interactive dashboards for policymakers and agricultural planners.

7. Scalability and Cloud Deployment

Deploy the system on cloud platforms (AWS, GCP, Azure) for large-scale use.

Support big datasets covering multiple regions or countries.

8. Integration with IoT Sensors

Connect IoT-based soil sensors for continuous monitoring of soil properties.

Enable real-time alerts and fertility monitoring.

9. Economic and Environmental Impact Analysis

Extend the system to estimate cost savings for farmers and environmental benefits from optimized fertilizer usage.

Provide sustainable farming recommendations.

5.4 Conclusion

This study developed a Soil Fertility Index (SFI) system that combines Machine Learning and GIS to provide real-time, location-specific insights on soil fertility in the Sunyani Municipality. Using 142 soil samples and a Random Forest Regressor model, the system achieved strong performance with 95% training accuracy and 87% validation accuracy, supported by low error margins.

The project successfully produced a framework that generates continuous fertility scores, classifies soils into categories (Excellent, Good, Average, Poor), and displays them through interactive maps. The web-based platform makes predictions accessible to farmers, researchers, and policymakers, supporting better decision-making in agriculture.

The study also contributes to sustainable development by aligning with SDG 2 (Zero Hunger), SDG 13 (Climate Action), and SDG 15 (Life on Land). It bridges computer science, GIS, and agriculture, showing how digital tools can improve food security and land management.

Future improvements should include expanding the dataset, testing advanced models like Gradient Boosting and Deep Learning, adding mobile access with offline features, and integrating IoT sensors and cloud deployment for continuous monitoring.

In summary, this work demonstrates that machine learning combined with GIS provides a powerful, scalable, and cost-effective solution for soil fertility monitoring, with the potential to transform agricultural productivity and sustainability in Ghana and beyond.

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